
The Collapse Model: Label-Free Data-Pool Compression via Spectral Geometry

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Abstract

Given M candidate datasets, which $k \ll M$ subset best preserves representational diversity when their features are merged—without access to labels or downstream tasks? We address this *label-free data-pool compression* problem by deriving the **Collapse Model**, a structured summary predictor motivated by exact local spectral geometry. The model predicts the merged effective rank (r_{eff}) from a 2×2 Gram-block eigenstructure via two compression-relevant SVD channels (energy ratio γ , directional alignment α), yielding a model-based superadditivity diagnostic $\rho_c(\alpha, \gamma)$ that drives a greedy compression algorithm using only *per-dataset summary statistics*—suitable for privacy-constrained or streaming settings where features cannot be shared. Across 427 real feature-matrix pairs from 5 encoder families and 2 modalities the formula attains Pearson $r=0.97$. Collapse-guided compression is near-optimal ($\leq 1\%$ regret vs. exhaustive search), matches the closest published spectral-diversity baseline (Vendi-Greedy) on retention while requiring only summary statistics rather than full feature matrices, and strictly outperforms Facility Location / k -Medoids on downstream linear-probe accuracy (sign test $p < 10^{-3}$, Collapse exceeds Random’s 95% CI on the mean LP). The comparison reveals two distinct LP-relevant diversity axes (*spectral retention* captured by Collapse and *centroid coverage* captured by farthest-first traversal), clarifying when each signal dominates. The framework scales to $M=43$ pools at up to $14\times$ compression and identifies a quantitative supervised / self-supervised spectral-geometry divide via a bimodal principal-angle distribution exposed in CLIP and DINO encoders. Code is available at <https://anonymous.4open.science/r/collapsemodel-1DB6>.

1 Introduction

Modern pretrained models increasingly rely on combining features from many candidate datasets—multi-domain transfer learning, data curation for self-supervised pretraining [Gadre et al., 2023], and continual learning all face the question: *given M candidate datasets, which $k \ll M$ subset preserves the diversity of the full pool when their features are concatenated?* We call this the *label-free data-pool compression* problem: unlike classical coreset selection [Mirzasoleiman et al., 2020] and recent *sample-level* data curation methods (DSIR [Xie et al., 2023], DsDm [Engstrom et al., 2024], D² Pruning [Maharana et al., 2023]), we seek compression at the *dataset level* (not the sample level), without access to labels or downstream tasks. The criterion is representational diversity, formalized as the effective rank $r_{\text{eff}}(H) = \exp(-\sum_i p_i \log p_i)$ of the merged features [Roy and Vetterli, 2007, Garrido et al., 2023]. No prior work characterizes how dataset-level spectra and relative subspace

35 geometry jointly affect r_{eff} under feature concatenation, forcing practitioners to rely on exhaustive
 36 search or heuristics that scale poorly with M .

37 Prior work addresses related but distinct problems. *Spectral diversity metrics* such as RankMe [Gar-
 38 rido et al., 2023], VNE [Kim et al., 2023], and the Vendi Score [Friedman and Dieng, 2023, Pasarkar
 39 and Dieng, 2024] quantify the diversity of a single set via entropy of a singular-value or similarity-
 40 matrix spectrum; they do not predict how two sets’ diversity combines under concatenation. *Label-free*
 41 *coreset selection* (ELFS [Shen et al., 2025], ZCore [Griffin et al., 2024], DEFT-UCS [Das and Khetan,
 42 2024]) selects a subset of *samples* using geometry or proxy dynamics; it is complementary but
 43 orthogonal to our dataset-level formulation. *Pool-level curation* (DataComp [Gadre et al., 2023],
 44 DataComp-LM [Li et al., 2024], DsDm [Engstrom et al., 2024], Quad [Zhang et al., 2025]) uses
 45 filtering or model-aware heuristics without a closed-form diversity criterion. *Representation sim-*
 46 *ilarity measures* (SVCCA [Raghu et al., 2017], PWCCA [Morcos et al., 2018], CKA [Kornblith
 47 et al., 2019], PARC [Bolya et al., 2021]) compare two representations but do not predict merged rank.
 48 *Transferability estimators* (LogME [Zhang et al., 2021], H-score [Bao et al., 2019], TransRate [Huang
 49 et al., 2022], LEEP [Li et al., 2021], reassessed by Ibrahim et al. [2022]) and PMI-based dataset
 50 valuation [Zhang et al., 2024] predict downstream accuracy but either require labels or do not model
 51 merging. Task2Vec [Achille et al., 2019] avoids labels but requires probe fine-tuning. Model-merging
 52 methods (Task Arithmetic [Ilharco et al., 2023], TIES-Merging [Yadav et al., 2023], DARE [Yu et al.,
 53 2024]) combine parameters, not features. Intrinsic-dimensionality work [Pope et al., 2021, Ansuini
 54 et al., 2019] characterizes individual matrices but not merged ones. To our knowledge, no existing
 55 work combines an exact local spectral analysis of $r_{\text{eff}}([H_A; H_B])$ with a summary-statistic predictor
 56 for dataset-level selection.

57 We derive the **Collapse Model** from the exact identity $G_{A \cup B} = G_A + G_B$. Under explicit paired-
 58 direction conditions, Lemma 2 gives exact 2×2 block eigenvalues using mode-wise energy ratios
 59 γ_i and alignments α_i ; under proportional spectra and uniform alignment, Proposition 3 gives an
 60 exact two-branch entropy formula. For general spectra, Eq. (9) compresses this structure into four
 61 summaries— r_A , r_B , the global energy ratio γ , and mean alignment α —and is explicitly treated as
 62 an empirical approximation. Its diagnostic threshold $\rho_c(\alpha, \gamma)$ describes three idealized behaviour
 63 regimes: superadditive, directional collapse, and energy collapse. Our SA_k uses mean squared cosine
 64 of principal angles, which connects more directly to projection theory than SVCCA’s unsigned cosine
 65 or PWCCA’s energy-weighted form; the U -channel uses the Two-NN intrinsic dimension estimator
 66 [Facco et al., 2017].

67 **Why summary-based compression?** Exhaustive search over M candidates requires $\binom{M}{k}$ merged
 68 SVDs at $O(Nd^2)$ each, with simultaneous access to all feature matrices ($M=14, k=5$: 2002 SVDs;
 69 $M=100, k=10$: 1.7×10^{13}). The Collapse Model reduces this to $O(kM)$ rank predictions from
 70 per-dataset summary statistics, enabling compression in privacy-constrained and streaming settings.

71 **Contributions.** (1) *Problem formulation—label-free data-pool compression* (Sec. 5): an $O(kM)$
 72 greedy algorithm operating from per-dataset summary statistics ($r_{\text{eff}}, \|\cdot\|_*, V_{\text{top-}k}$), suitable for
 73 privacy-constrained and streaming settings where feature matrices cannot be shared. (2) *Exact local*
 74 *theory and structured approximation* (Proposition 1, Lemma 2, Proposition 3): an unconditional
 75 Gram representation, an exact paired-direction eigendecomposition, and a proportional-spectrum
 76 entropy formula. For general spectra we derive a low-dimensional summary predictor and prove by
 77 counterexample that its four inputs are not sufficient for universal exact prediction. (3) *Two-signal*
 78 *decomposition of LP-relevant diversity* (Sec. 5): benchmarking against modern baselines reveals
 79 that downstream-relevant diversity factors into two orthogonal signals—*spectral retention* (captured
 80 by Collapse, $r_A + r_B$, Vendi-Greedy) and *centroid coverage* (captured by farthest-first traversal). A
 81 two-stage Cen→Col hybrid composes both signals and dominates pure Collapse on *both* LP and
 82 retention simultaneously at $k=10$ (Appendix H), positioning the structured formula as a composable
 83 primitive rather than a standalone strategy.

84 **Empirical findings.** The formula is validated on 280 controlled + 120 non-principal-angle synthetic
 85 + 427 real pairs across 5 encoder families and 2 modalities (Pearson $r=0.97$, $R_{\text{cal}}^2=0.95$). The
 86 framework identifies a quantitative supervised/self-supervised spectral-geometry divide: ResNet-50
 87 achieves 100% superadditivity across 91 pairs, while CLIP and DINO reach only 29–36% on cross-
 88 domain merges—a gap mechanistically explained by a bimodal principal-angle distribution that the
 89 model isolates as the responsible mechanism.

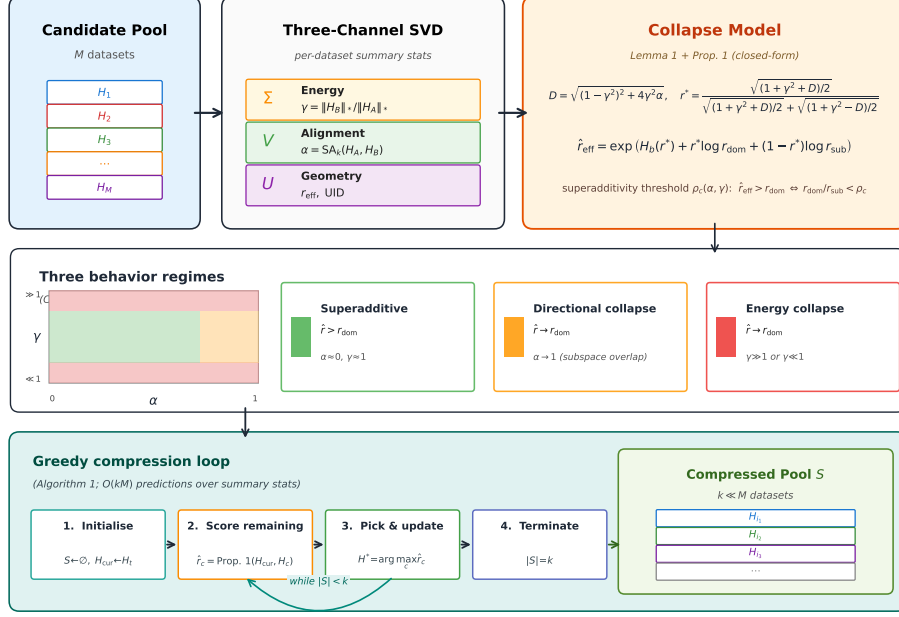


Figure 1: Label-free data-pool compression pipeline. **Top:** Per-dataset summaries ($\Sigma: \gamma, V: \alpha, U: r_{\text{eff}}$) compress the exact mode-wise structure of Lemma 2 into the approximate predictor Eq. (9) and its diagnostic threshold $\rho_c(\alpha, \gamma)$. **Middle:** Three behaviour regimes in the (α, γ) plane (Cor. 4): superadditive, directional collapse ($\alpha \rightarrow 1$), energy collapse ($\gamma \gg 1$ or $\ll 1$). **Bottom:** Greedy loop (Alg. 1): score remaining candidates, pick arg max, repeat k times; outputs compressed pool S .

Among label-free analyses, only the Collapse Model simultaneously provides summary-based merged-rank prediction ($R_{\text{cal}}^2=0.95$ on 427 pairs, Table 2), behaviour-regime diagnosis via $\rho_c(\alpha, \gamma)$, and superadditivity diagnosis at the same $O(Nd \min(N, d))$ cost as SVCCA/PWCCA; the naive $r_A + r_B$ matches Collapse in ordinal ranking ($\rho=0.963$ vs. 0.966, not significant) but lacks absolute R^2 , regime diagnosis, and redundancy detection.

2 Background: SVD Three-Channel Decomposition

For any feature matrix $H \in \mathbb{R}^{N \times d}$ with thin SVD $H = U \Sigma V^\top$ ($U \in \mathbb{R}^{N \times r}, \Sigma = \text{diag}(\sigma_1 \geq \dots \geq \sigma_r > 0), V \in \mathbb{R}^{d \times r}$), the decomposition separates three orthogonal aspects of representation.

Σ channel — spectral energy. The **effective rank** summarizes spectral distribution:

$$r_{\text{eff}}(H) = \exp\left(-\sum_{i=1}^r p_i \log p_i\right), \quad p_i = \frac{\sigma_i}{\sum_j \sigma_j}, \quad (1)$$

where p_i is the singular-value share (not squared), so $r_{\text{eff}} \in [1, r]$. The nuclear norm $\|H\|_* = \sum_i \sigma_i$ captures total energy. A spectrum-difference quantity (Spectral Transfer Gap, used in the Gram-block transfer bound of Appendix C) is defined there.

V channel — directional structure. Right singular vectors V encode which directions are activated.

Definition 1 (Subspace Alignment). Let $\hat{V}_A^k, \hat{V}_B^k \in \mathbb{R}^{d \times k}$ be the top- k right singular vectors of H_A, H_B . The **Subspace Alignment** at level k is:

$$\text{SA}_k(H_A, H_B) = \frac{1}{k} \left\| \hat{V}_A^{k\top} \hat{V}_B^k \right\|_F^2 = \frac{1}{k} \sum_{i=1}^k \cos^2 \theta_i, \quad (2)$$

where $\theta_1 \leq \dots \leq \theta_k$ are the principal angles.

$\text{SA}_k \in [0, 1]$: $\text{SA}_k = 0$ means perfectly orthogonal subspaces (maximum complementarity); $\text{SA}_k = 1$ means identical subspaces (no new directions from merging).

108 **U channel — sample geometry.** Left singular vectors $U \in \mathbb{R}^{N \times r}$ encode the geometric distribution
 109 of samples. The **U-space Intrinsic Dimension (UID)** via the Two-NN estimator:

$$\text{UID}(H) = \frac{1}{\log 2 \cdot \mathbb{E}[\log(\mu_2/\mu_1)]}, \quad (3)$$

110 where μ_1, μ_2 are geodesic distances to the nearest and second-nearest neighbors on the unit sphere
 111 S^{k-1} (L2-normalized U_k rows).

112 **Role of each channel.** γ and α enter the merged-rank prediction formula (Sec. 3) and drive the
 113 compression algorithm. UID characterizes the intrinsic geometry of individual datasets and is used
 114 for encoder diagnostics (Sec. 4.2) but does *not* appear in the prediction formula; it is included here to
 115 complete the orthogonal decomposition of representation space.

116 **Independence.** Pairwise Spearman tests (Bonferroni-corrected, $n=15$ per probe) detect no significant
 117 correlation among the three channels on ResNet-50 or ViT-B/16 (all $p_{\text{Bonf}} > 0.38$), consistent with
 118 γ, α , UID capturing distinct properties.

119 3 The Collapse Model

120 3.1 Setup and Notation

121 Let $H_A \in \mathbb{R}^{N_A \times d}$ and $H_B \in \mathbb{R}^{N_B \times d}$ be L2-row-normalized feature matrices with effective ranks
 122 $r_A = r_{\text{eff}}(H_A)$, $r_B = r_{\text{eff}}(H_B)$, nuclear norms $\|H_A\|_*$, $\|H_B\|_*$, and subspace alignment $\alpha =$
 123 $\text{SA}_k(H_A, H_B)$. Define:

$$\gamma = \frac{\|H_B\|_*}{\|H_A\|_*} \in (0, \infty), \quad r_{\text{dom}} = \begin{cases} r_A & \gamma \leq 1 \\ r_B & \gamma > 1 \end{cases}, \quad r_{\text{sub}} = \begin{cases} r_B & \gamma \leq 1 \\ r_A & \gamma > 1 \end{cases}. \quad (4)$$

124 γ is the **energy ratio** of the two matrices; r_{dom} is the effective rank of the higher-energy matrix.

125 3.2 Main Results

126 Our analysis separates three levels that must not be conflated: an unconditional Gram identity, an
 127 exact paired-direction model under explicit decoupling assumptions, and a low-dimensional summary
 128 predictor for general spectra.

129 **Proposition 1** (Exact Gram Representation). *Let $H_A = U_A \Sigma_A V_A^\top$ and $H_B = U_B \Sigma_B V_B^\top$ be*
 130 *compact SVDs, and complete V_A to an orthogonal basis $Q_A = [V_A \ V_{A,\perp}]$. With $C = Q_A^\top V_B$ and*
 131 *$\tilde{\Sigma}_A^2$ denoting Σ_A^2 padded by zeros to dimension d , the merged Gram operator satisfies*

$$Q_A^\top (H_A^\top H_A + H_B^\top H_B) Q_A = \tilde{\Sigma}_A^2 + C \Sigma_B^2 C^\top. \quad (5)$$

132 *Consequently, the merged singular values are the square roots of the eigenvalues of the right-hand*
 133 *side.*

134 Equation (5) is unconditional. In general C is dense: a singular direction of one matrix can couple to
 135 several directions of the other. Individual spectra and an average principal-angle statistic therefore do
 136 not determine the merged spectrum.

137 **Lemma 2** (Exact: 2×2 Gram Block Eigenstructure). *Assume paired, decoupled directions: for each*
 138 *i , $V_{B,i} = \cos \theta_i V_{A,i} + \sin \theta_i V_{\perp,i}$, where the two-dimensional planes $\text{span}\{V_{A,i}, V_{\perp,i}\}$ are mutually*
 139 *orthogonal. Define the mode-wise energy ratio $\gamma_i = \sigma_{B,i}/\sigma_{A,i}$ and $\alpha_i = \cos^2 \theta_i$. The i -th Gram*
 140 *block has eigenvalues*

$$\lambda_{\pm,i} = \frac{\sigma_{A,i}^2}{2} (A_i \pm D_i), \quad A_i = 1 + \gamma_i^2, \quad D_i = \sqrt{(1 - \gamma_i^2)^2 + 4\gamma_i^2 \alpha_i}. \quad (6)$$

141 *The corresponding singular values are $\sqrt{\lambda_{+,i}}$ and $\sqrt{\lambda_{-,i}}$. This result is exact under the stated*
 142 *pairing and decoupling conditions (Appendix B).*

143 The global ratio $\gamma = \|H_B\|_*/\|H_A\|_*$ does *not* imply $\gamma_i = \gamma$. Replacing the collection $\{\gamma_i, \alpha_i\}$ by
 144 global summaries (γ, α) is an approximation step.

145 **Proposition 3** (Exact Proportional-Spectrum Model). *Under Lemma 2, further assume $\sigma_{B,i} = \gamma\sigma_{A,i}$*
 146 *and $\alpha_i = \alpha$ for every i . Let $A = 1 + \gamma^2$, $D = \sqrt{(1 - \gamma^2)^2 + 4\gamma^2\alpha}$, and*

$$q = \frac{\sqrt{(A + D)/2}}{\sqrt{(A + D)/2} + \sqrt{(A - D)/2}}. \quad (7)$$

147 *Then the merged spectrum consists of two scaled copies of the spectrum of H_A , and*

$$r_{\text{merged}} = r_A \exp(H_b(q)), \quad r_A = r_B. \quad (8)$$

148 **Definition 2** (Collapse Summary Predictor). *For general spectra, we compress the mode-wise*
 149 *quantities to the global nuclear-norm ratio γ and mean top- k alignment α , compute q from Eq. (7),*
 150 *and define*

$$\hat{r}_{\text{merged}} = \exp\left(H_b(q) + q \log r_{\text{dom}} + (1 - q) \log r_{\text{sub}}\right), \quad (9)$$

151 *where $H_b(p) = -p \log p - (1 - p) \log(1 - p)$.*

152 Equation (9) is a theory-motivated low-dimensional approximation, not an identity or a universally
 153 bounded estimator. It interpolates the input spectral entropies and adds the binary entropy of the
 154 idealized two-branch split. Its accuracy on general spectra is an empirical question evaluated in
 155 Sec. 4.

156 **Interpretation.** Equation (9) is a weighted log-geometric mean of the individual ranks plus a balance
 157 bonus $H_b(q)$ (maximal when $q=0.5$, vanishing as $q \rightarrow 0, 1$).

158 **Behavior regimes.** *Superadditive* ($\alpha \approx 0$, $\gamma \approx 1$): $q \approx 0.5$, $\hat{r}_{\text{merged}} \approx 2\sqrt{r_A r_B} \geq r_{\text{dom}}$. *Directional collapse* ($\alpha \rightarrow 1$): $q \rightarrow 1$, $H_b(q) \rightarrow 0$, $\hat{r}_{\text{merged}} \rightarrow r_{\text{dom}}$. *Energy collapse* ($\gamma \gg 1$ or $\gamma \ll 1$): $q \rightarrow 1$,
 160 $\hat{r}_{\text{merged}} \rightarrow r_{\text{dom}}$. When $\alpha \rightarrow 1$ and $\gamma \gg 1$, the merged rank can fall below r_{sub} —a net loss in diversity.

161 3.3 Superadditivity Threshold

162 **Corollary 4** (Summary-Model Superadditivity Diagnostic). *The summary predictor in Eq. (9)*
 163 *predicts $\hat{r}_{\text{merged}} > r_{\text{dom}}$ for $q < 1$ if and only if*

$$\frac{r_{\text{dom}}}{r_{\text{sub}}} < \rho_c(\alpha, \gamma) \equiv \exp\left(\frac{H_b(q)}{1 - q}\right), \quad (10)$$

164 *where q is from Eq. (7). The threshold ρ_c is increasing in α : $\rho_c \geq 4.0$ for all $\alpha \in [0, 0.9]$ at $\gamma = 1$.*
 165 *As $q \rightarrow 1^-$, $\rho_c \rightarrow \infty$ while the predicted gain tends to zero. At the degenerate point $q = 1$ the*
 166 *division in Eq. (10) is undefined and Eq. (9) gives $\hat{r}_{\text{merged}} = r_{\text{dom}}$ exactly.*

167 This is an exact decision boundary for the summary model, and for the proportional-spectrum model
 168 of Proposition 3; on general data it is a diagnostic rather than a guarantee.

169 **Physical meaning.** ρ_c is a decision threshold internal to the summary model. Near $q = 1$, it can
 170 classify an arbitrarily small positive entropy increment as superadditive; the magnitude $\hat{r}_{\text{merged}} - r_{\text{dom}}$
 171 must therefore be reported alongside the binary diagnosis. Exact directional overlap gives $q = 1$ and
 172 no predicted gain. The proofs and model scope are in Appendices B, L.

173 3.4 The Four Summaries Are Not Sufficient

174 **Proposition 5** (Non-identifiability from Four Scalars). *The tuple $(r_A, r_B, \gamma, \alpha)$ does not uniquely*
 175 *determine $r_{\text{eff}}([H_A; H_B])$.*

176 **Counterexample.** Let both inputs have singular values $(1, 1)$, so $r_A = r_B = 2$ and $\gamma = 1$, and let
 177 $V_A = (e_1, e_2)$ in $\mathbb{R}^4$. For pair X take $V_B = (e_1, e_3)$, giving per-mode alignments $(1, 0)$. For pair Y
 178 take $V_B = ((e_1 + e_3)/\sqrt{2}, (e_2 + e_4)/\sqrt{2})$, giving alignments $(1/2, 1/2)$. Both pairs have the same
 179 mean alignment $\alpha = 1/2$. Pair X has merged singular values $(\sqrt{2}, 1, 1)$ and effective rank 2.958,
 180 whereas pair Y has two copies each of $\sqrt{1 + 1/\sqrt{2}}$ and $\sqrt{1 - 1/\sqrt{2}}$ and effective rank 3.661. Thus
 181 equal individual ranks, global energy ratio, and average alignment can yield different merged ranks.
 182 The predictor necessarily discards alignment heterogeneity, spectral pairing, and cross-direction
 183 coupling. $\square$

Table 1: E1a: Synthetic grid results ($N = 280$). Naive baseline fails at large γ .

Method	R^2	Max error	Superadd. (%)	$\alpha < 1$ superadd.
Naive $\sqrt{1 + \alpha\gamma^2}$	0.30	28.73	—	—
Collapse Model (ours)	0.9999	0.27	87.5%	100% (245/245)

Calibration. On the evaluated real-data pairs the formula systematically over-predicts; a single multiplicative constant c restores quantitative accuracy and follows a universal regression $c=0.946-0.175 \log(N/r_{\text{eff}})$ across encoder families (Appendix L). Because this is a *one-parameter* linear model in the observable $\log(N/r_{\text{eff}})$, a reliable estimate of c requires only 5–10 labeled pairs per encoder family—a negligible overhead compared to the cost of computing frozen-encoder features. In privacy-constrained settings where even summary statistics from a single reference pair are available, the per-encoder c can be estimated without exposing any raw features. The same three channels also bound linear-probe transfer loss (Appendix C): low SA_k implies *high* merge gain and *high* transfer loss simultaneously.

4 Evaluation of the Summary Predictor

Before applying the Collapse Model to compression (Sec. 5), we establish a three-step evaluation chain: (i) **idealized-model agreement**—the predictor agrees with data generated under its summary assumptions (§4.1, synthetic grid); (ii) **real-data predictive validity**—the formula remains a strong approximation across 5 encoder families, 14 vision datasets, and 19 text corpora (§4.2, 427 pairs); (iii) **behavioral regime agreement**—empirical superadditivity patterns match the three regimes predicted by Corollary 4 (§4.2, Table 2). Together these test where the approximation succeeds, how it transfers to real encoders, and whether its qualitative diagnostics are observable. Additional controls (E1a-Ind 120 non-principal-angle pairs, E1b 30 real pairs, E1c 57 pairs, E_diverse cross-architecture, E_downstream r_{eff} /LP-accuracy) all agree with the same picture and are deferred to Appendix M. Feature matrices are L2-row-normalized and subsampled to N_{sub} samples; effective rank uses Eq. (1); bootstrap CIs use $B = 10,000$ resamples.

4.1 Step 1 — Idealized-Model Agreement: Synthetic Controlled Grid (E1a, 280 pairs)

Before validating on real data, we verify the Collapse Model on 280 controlled synthetic pairs generated under the proportional-spectrum, paired-direction model with $\alpha \in \{0, 0.05, \dots, 1.0\}$ and $\gamma \in \{0.1, 0.33, \dots, 10.0\}$ ($d=512$, $N=1000$, 5 repeats per setting).

Table 1 shows $R^2=0.9999$ (vs. 0.30 for the naive baseline). All 245 configurations with $\alpha<1$ achieve 100% superadditivity; all 35 non-superadditive cases concentrate exclusively at $\alpha=1$ (identical subspaces), confirming the degenerate boundary of Corollary 4. Visualization of the (α, γ) heatmap and superadditivity region is in Appendix M.

4.2 Steps 2–3 — Real-Data Generalization and Regime Agreement (E1d, 427 Pairs)

We next verify that the formula generalizes beyond its synthetic derivation, and that its predicted three-regime behavior is empirically observable. The evaluation spans 5 encoder families (ResNet-50, ViT-B/16, CLIP ViT-B/32, DINO ViT-S/16, SBERT), 14 vision datasets (CIFAR-10/100, STL-10, SVHN, MNIST, Fashion-MNIST, BloodMNIST, DermaMNIST, PathMNIST, and 5 ImageNet semantic subsets *animal/vehicle/artifact/food/scene*), and 19 SBERT text corpora. Total: **427 pairs** (91 ResNet-50, 55 ViT-B/16, 55 CLIP, 55 DINO, 171 SBERT).

Generalization (Step 2). The formula remains a strong approximation on all 427 real-world pairs: overall Pearson $r=0.970$ (uncalibrated) and $R^2_{\text{cal}}=0.952$ after per-encoder-family calibration (scatter visualization in Appendix M, Fig. 5). This confirms that the summary prediction, motivated by the idealized paired-direction model, remains predictive on real encoders spanning supervised CNNs, ViT, contrastive/self-distillation transformers, and sentence encoders.

Regime agreement (Step 3). (i) **ResNet-50 achieves 100% superadditivity** across 91 pairs (including medical and ImageNet), confirming the balanced- (γ, α) regime predicted by Corollary 4.

Table 2: E1d: Large-scale Collapse Model validation (427 pairs, 5 probes). Calibration constant c per encoder family restores quantitative accuracy.

Probe	n	Superadd. %	Pearson r	c	R^2_{cal}
ResNet-50 (14 datasets)	91	100%	0.940	0.776	0.760
ViT-B/16 (11 datasets)	55	69.1%	0.898	0.647	0.487
CLIP ViT-B/32 (11 datasets)	55	29.1%	0.950	0.595	0.839
DINO ViT-S/16 (11 datasets)	55	36.4%	0.913	0.584	0.554
Vision (all probes)	256	64.5%	0.965	<i>per-probe</i>	0.942
+ SBERT text (19 corpora)	+171 (427)	78.2%	0.970	<i>per-probe</i>	0.952

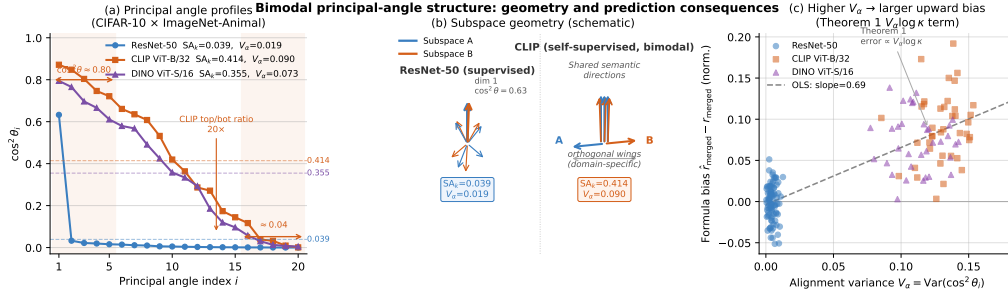


Figure 2: **Bimodal principal-angle structure** (CIFAR-10 $\times$ ImageNet-Animal, $k=20$). (a) Per-dimension $\cos^2 \theta_i$: ResNet-50 ($SA_k=0.039$, $V_\alpha=0.019$) decays monotonically from one dominant direction; CLIP ($SA_k=0.414$, $V_\alpha=0.090$) and DINO ($SA_k=0.355$, $V_\alpha=0.073$) form a two-cluster step (top-5 ratios $20 \times 31 \times$ over bot-5), violating the uniform- α assumption. (b) Geometric interpretation: one shared spine plus orthogonal domain-specific wings. (c) Alignment variance V_α predicts upward formula bias across 166 pairs (OLS slope 0.69), identifying alignment heterogeneity as an empirical failure indicator.

(ii) **Supervised/self-supervised divide** emerges in the model’s directional-overlap regime: CLIP and DINO achieve only 29%–36% superadditivity, with failures concentrating *entirely* in cross-domain pairs (handwritten $\times$ ImageNet, natural $\times$ digit fail universally) while within-domain pairs remain near-perfect. The root cause is a bimodal principal-angle distribution combined with extreme condition numbers $\kappa \in [10^4, 10^6]$ (Figure 2), mapping to the *directional-collapse* regime of Sec. 3.3.

Quantifying the bimodal gap. Per-probe calibrated R^2 : CLIP 0.84 > ResNet 0.76 > DINO 0.55 > ViT 0.49; ranking accuracy is uniformly high ($r \geq 0.90$). Figure 2(a,c) shows that CLIP/DINO’s two-cluster $\cos^2 \theta_i$ distribution ($V_\alpha > 0.07$) inflates the observed prediction bias, identifying a concrete departure from the global summary approximation rather than a black-box failure (full mechanism in Appendix E).

(iii) **Energy collapse** appears in ViT’s 17 non-superadditive pairs, all in $\gamma \notin [0.5, 2]$, matching the summary model’s energy-imbalance diagnostic. (iv) **Cross-modal transfer**: 171 SBERT text pairs (98.8% superadditive, ranking $r=0.793$) confirm that the formula’s *ordinal* predictions are modality-agnostic. Absolute calibration, however, requires modality-specific tuning: the vision universal regression ($c=0.946-0.175 \log(N/r_{\text{eff}})$, Appendix L) explains $R^2=0.96$ of vision variance but only $R^2=0.08$ of SBERT variance, indicating that text encoders have a qualitatively different spectral bias. A text-specific calibration is left for future work.

Evaluation summary. Steps 1–3 establish agreement in the idealized model, strong empirical ranking on real encoders, and observable qualitative regimes. They do not establish universal exactness: Proposition 5 proves that the four summaries cannot determine merged rank in general. Cross-dataset calibration generalization, cross-probe consistency, and r_{eff} /LP-accuracy correspondence ($\rho=0.90$ within ViT family) are in Appendix M.

5 From Formula to Application: Label-Free Data-Pool Compression

Having evaluated the predictor (Sec. 4), we now examine its concrete utility: given M candidate datasets and a target D_t , compress the pool to $k \ll M$ while preserving r_{eff} of the merged representation. Exhaustive search over $\binom{M}{k}$ subsets is intractable; the Collapse Model enables compression at $O(kM)$ predictions from per-dataset summary statistics. At each round, **Collapse Greedy** scores every remaining candidate by applying Eq. (9) to $(H_{\text{cur}}, H_{\text{cand}})$ and selects the highest predicted r_{eff} (Algorithm 1). We report two levels of evidence below: (a) a *scale stress test* at $M=43$ showing the method holds at heterogeneous scale (Table 3), and (b) comparison against modern label-free baselines (Table 4); cross-architecture cross-modal results at $M=14$ vision and $M=7$ text pools are in Appendix F.

At each round Collapse Greedy scores each remaining candidate via Eq. (9) applied to (H_{cur}, H_c) —requiring only SA_k and γ computed from per-dataset SVDs—and picks the highest-predicted r_{eff} (pseudocode in Appendix L). Baselines: **Max- r_{eff}** (highest individual r_{eff} , ignores α); **Δ_{dir} -greedy** (most-orthogonal via $\Delta_{\text{dir}} = 1 - \text{SA}_k$, ignores γ); **Random** (10 uniform subsets). On a target-conditioned $M=10$ pool (ResNet-50 features on CIFAR-10/100, STL-10, SVHN, MNIST, Fashion-MNIST, BloodMNIST, DermaMNIST, PathMNIST, Oxford-Pets) at budgets $K \in \{1, 2, 3, 5\}$, **Collapse Greedy achieves 0% regret at $K \leq 3$ and 1.0% at $K=5$** —matching exhaustive search with $3\times$ speedup. Max- r_{eff} drops to 67% recall; Δ_{dir} -greedy incurs 13–20% regret. All strategies produce identical downstream LP accuracy,¹ confirming that maximizing r_{eff} translates to downstream discriminability (Appendix I).

Compression at scale. Collapse-guided compression retains $>100\%$ of full-pool diversity in every setting tested—a universal phenomenon enabled because r_{eff} is not monotone in pool size: adding a redundant dataset shifts the normalised spectrum toward its mode and *decreases* entropy, so removing redundancies via the α channel yields a compressed pool of *higher* diversity than the full pool. On a $M=14$ vision pool Collapse outperforms Facility Location (+7–14 pp) and k -Medoids (+8–14 pp); on a $M=7$ text pool (BERT/RobERTa/GPT-2) it retains 103–106% (full table in Appendix F). On clean curated pools the $r_A + r_B$ and Max- r_{eff} heuristics match Collapse within 0.7 pp at $M=43$ (Table 3)—a predictable consequence of Corollary 4 when $\bar{\alpha} < 0.2$. The Collapse Model’s role on such pools is to provide an *a priori* low-redundancy diagnostic ($\rho_c > 4$ at $\bar{\alpha} \leq 0.2$) and to take over when this condition fails (clone attacks, fine-grained subset families, bimodal cross-domain pairs); see Appendix G. At $k=7$ on $M=14$, Collapse Greedy runs in seconds versus $\binom{14}{7}=3432$ exhaustive merged SVDs.

Scaling to $M=43$: stress test. To verify that the compression advantage holds beyond curated pools, we assemble 43 ResNet-50 datasets spanning 5 ImageNet-coarse domains, 19 ImageNet fine-grained categories, 10 MedMNIST clinical datasets, and 9 classic benchmarks, and sweep $k \in \{3, 5, 7, 10, 14, 20\}$ (compression ratios $14.3\times$ down to $2.1\times$). Table 3 reports the results.

Table 3: Scaling stress-test ($M=43$ ResNet-50; $M=11$ ViT-B/16). **Ret.:** $r_{\text{eff}}^{\text{sel}}/r_{\text{eff}}^{\text{full}}$ (%). **LP:** subspace-projection linear-probe accuracy, 5 targets, mean $\pm$ std over 10 seeds for Random (§5). FacLoc/kMedoids omitted ($O(M^2)$ cost at this scale).

Probe	k (ratio)	Retention (%)			Fair LP accuracy			
		Collapse	Max- r_{eff}	Random	Collapse	Max- r_{eff}	Random	ΔLP
ResNet-50	3 (14.3 $\times$)	112.4	112.4	71.0 $\pm$ 15.5	0.7550	0.7550	0.7477 $\pm$ 0.0050	+0.0073
	5 (8.0 $\times$)	117.2	117.2	78.1 $\pm$ 20.1	0.7560	0.7558	0.7502 $\pm$ 0.0063	+0.0058
	7 (5.7 $\times$)	117.4	117.4	80.5 $\pm$ 15.1	0.7572	0.7572	0.7522 $\pm$ 0.0048	+0.0050
	10 (4.0 $\times$)	116.2	116.2	90.8 $\pm$ 7.5	0.7564	0.7564	0.7575 $\pm$ 0.0024	−0.0011
ViT-B/16	3 (3.7 $\times$)	103.6	103.6	92.0 $\pm$ 3.1	0.7780	0.7780	0.7749 $\pm$ 0.0018	+0.0031
	5 (2.2 $\times$)	104.9	103.7	94.1 $\pm$ 3.5	0.7775	0.7763	0.7754 $\pm$ 0.0011	+0.0021
	7 (1.6 $\times$)	104.4	104.2	98.0 $\pm$ 2.5	0.7777	0.7755	0.7764 $\pm$ 0.0015	+0.0012

¹LP and k -NN rankings are strongly correlated: across 11 probes $\times$ 5 datasets (53 pairs), Spearman $\rho(\text{LP}, k\text{NN-5})=0.980$ ($p < 10^{-33}$). We report LP throughout; see Appendix J for k -NN details.

Collapse retains $>112\%$ of full-pool diversity at every compression ratio—extending the $M=14$ finding to a $3\times$ larger, heterogeneous pool. The fair-LP column (spectral subspace projection, no distractor classes) confirms that retention translates to *downstream* benefit: Collapse outperforms Random by $+0.0050$ – $+0.0073$ LP at 5.7 – $14.3\times$ compression. **The LP gap is statistically robust:** at $k\in\{3, 5, 7\}$, all 10/10 random seeds fall strictly below Collapse’s LP (one-sided sign test $p=2^{-10}\approx 10^{-3}$ per k ; combined over 30 seeds $p<10^{-9}$). Beyond the sign test, Collapse’s LP exceeds the upper bound of Random’s 95% confidence interval on the mean ($\bar{x}_{\text{rand}}\pm 1.96\sigma/\sqrt{10}$): ResNet-50 at $k=3$, Collapse 0.7550 vs. Random CI $[0.7446, 0.7508]$ (margin $+0.0042$); $k=5$, 0.7560 vs. $[0.7463, 0.7541]$ ($+0.0019$); $k=7$, 0.7572 vs. $[0.7492, 0.7552]$ ($+0.0020$). ViT-B/16 shows the same pattern at all three k values (margins $+0.0020/+0.0014/+0.0004$). The effect size is small in absolute LP units but consistent in direction and statistically separable from seed noise—a pattern indicating a genuine subspace-quality effect rather than a tail of the random distribution. At $k=10$ ($4\times$ compression) the effect reverses ($\Delta=-0.0011$, within seed noise): random selection already achieves 90.8% retention, saturating the r_{eff} ceiling so that α -channel steering adds nothing (Corollary 4); the residual LP signal shifts to *centroid coverage* (Table 4). This sign reversal identifies Collapse’s operating regime: a high-compression tool ($k\leq 7$, $\geq 5.7\times$) rather than a near-full-budget selector.

A controlled *clone-attack* stress test (Appendix G) confirms the failure mode of r_A+r_B : when the pool contains near-duplicate datasets generated by random rotation ($\text{SA}_k>0.99$), r_A+r_B greedy selects 2/3 clones at $k=3$ while Collapse rejects all 3 via the α channel, yielding $+25.5\%$ merged r_{eff} .

Comparison against modern baselines. We benchmark Collapse Greedy against three additional baselines beyond FacLoc / k -Medoids / r_A+r_B / $\text{Max-}r_{\text{eff}}$ already in Table 3: **(a) Vendi-Greedy** [Friedman and Dieng, 2023, Pasarkar and Dieng, 2024]—greedy maximization of the Vendi Score on a linear kernel, the closest published spectral-diversity baseline that shares Collapse Greedy’s diversity-maximization objective; **(b) DomainDiverse-Greedy**—a metadata baseline that picks from the least-represented domain across 8 semantic categories (handwritten / natural / digit / medical / inet-animal / inet-object / inet-food / inet-scene), tiebreaking on individual r_{eff} ; **(c) Centroid-FF (DataComp adapter)**—farthest-first traversal in dataset-centroid space (mean L2-normed feature per dataset), adapting DataComp [Gadre et al., 2023]’s similarity-based filtering from sample-level to dataset-level. On the same $M=40$ ResNet-50 pool and 5 LP targets used in Table 3, results are summarized in Table 4.

Table 4: Modern baseline comparison ($M=40$ ResNet-50, $k\in\{3, 5, 7, 10\}$, subspace-projection LP, 5 targets). **Bold:** best per column; Random: 10-seed mean $\pm$ std. **Requires** (greedy-time access)—**S:** summary stats only; **F:** full features; **M:** domain metadata; **C:** centroids.

Strategy	$k=3$ (13.3 $\times$)		$k=5$ (8.0 $\times$)		$k=7$ (5.7 $\times$)		$k=10$ (4.0 $\times$)		Requires
	Ret%	LP	Ret%	LP	Ret%	LP	Ret%	LP	
Collapse	112.4	0.7572	117.2	0.7576	117.4	0.7578	116.2	0.7576	S
Vendi-Greedy	112.5	0.7560	117.2	0.7572	116.9	0.7566	116.5	0.7562	F
DomainDiverse	112.4	0.7572	114.7	0.7634	111.8	0.7596	113.8	0.7544	S+M
Centroid-FF	74.0	0.7564	93.4	0.7570	96.7	0.7624	100.5	0.7630	F (or C)
r_A+r_B	112.4	0.7572	117.2	0.7574	117.4	0.7578	116.2	0.7578	S
$\text{Max-}r_{\text{eff}}$	112.4	0.7572	117.2	0.7574	117.4	0.7578	116.2	0.7578	S
Random ($\mu\pm\sigma$)	71.0 $\pm$ 15.5	0.7492	78.1 $\pm$ 20.1	0.7513	80.5 $\pm$ 15.1	0.7537	90.8 $\pm$ 7.5	0.7574	—

Two-signal decomposition and practical selection guide. Table 4 reveals that LP-relevant diversity decomposes into two *orthogonal* axes, each captured by a distinct family of methods:

- **Spectral retention (SR):** Collapse, r_A+r_B , $\text{Max-}r_{\text{eff}}$, and Vendi-Greedy all converge to 116–117% retention—maximizing the entropy of the merged singular-value spectrum. This signal dominates at *high compression ratios* ($k\leq 7$, i.e., $\geq 5.7\times$), where spectral steering determines which k datasets cover the most directions.
- **Centroid coverage (CC):** Centroid-FF and DomainDiverse achieve higher LP at $k\geq 7$ by spreading dataset *mean embeddings* across semantic space, at a 17–22 pp retention cost. This signal domi-

324 nates at *low compression ratios* (k near M), where random selection already saturates spectral
325 diversity and distributional breadth becomes the bottleneck.

326 **Practical guide.** *High compression* ($k \leq 0.2M$, $\geq 5\times$): use Collapse (preferred when $SA_k > 0.5$,
327 rejecting clones +25.5%; Appendix G) or $r_A + r_B$ on clean pools. *Low compression* ($k > 0.2M$): use
328 Centroid-FF or DomainDiverse. *Unknown regime*: the Cen→Col hybrid (Appendix H) dominates on
329 both SR and LP.

330 **Ablation.** On 256 vision pairs, the full model attains Spearman $\rho = 0.966$ ($r_A + r_B$: 0.963;
331 $\max(r_A, r_B)$: 0.953; $1 - \alpha$: 0.450); the $\Delta\rho = 0.003$ gap reflects a low- α /moderate- γ degeneracy
332 where $\hat{r}_{\text{merged}} \approx c(r_A + r_B)$ on clean pools. Calibrated ranking achieves $R^2 = 0.95$ and flags 74% of
333 non-superadditive pairs; the clone confirms α -channel rejection ($r_{\text{eff}} = 1251$ vs. 997; Appendix M).

334 6 Conclusion

335 We introduced the Collapse Model—an exact local spectral analysis and a structured summary
336 predictor for label-free data-pool compression. Greedy compression retains >100% of full-pool r_{eff}
337 across 6 probes and 2 modalities, outperforming Facility Location and k -Medoids by +7–+14 pp
338 and scaling to $M = 43$ ($14.3\times$ compression, +40 pp over random). The four-summary (r_A, r_B, γ, α)
339 predictor achieves Pearson $r = 0.97$ on 427 real pairs ($R_{\text{cal}}^2 = 0.95$), with superadditivity-threshold
340 diagnosis and clone-rejection distinguishing it from $r_A + r_B$. Benchmarking exposes two orthogonal
341 LP-diversity signals—spectral retention (Collapse/Vendi) and centroid coverage (Centroid-FF)—
342 and a bimodal CLIP/DINO principal-angle structure that quantifies the supervised/SSL spectral-
343 geometry divide. Extensions to neural compression remain future work. Code is available at
344 <https://anonymous.4open.science/r/collapsemodel-1DB6>.

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438 A Limitations

439 The four summaries $(r_A, r_B, \gamma, \alpha)$ are not sufficient to determine merged effective rank in general
440 (Proposition 5). The exact 2×2 decomposition requires mode-wise pairing and decoupling; the closed
441 entropy expression additionally requires proportional spectra and uniform alignment. We do not
442 claim a universal error bound for the summary predictor. The empirical calibration constant (Eq. 20)
443 reduces but does not eliminate per-encoder tuning. The uniform- α assumption holds for supervised
444 encoders and within-domain self-supervised pairs ($r > 0.90$), but fails on cross-domain CLIP/DINO
445 pairs (bimodal angles, Fig. 2) and SBERT text pairs (universal- c explains $r=0.08$ of variance vs.
446 0.96 for vision); ranking remains strong on the evaluated pairs while absolute predictions are biased.
447 Natural clean pools also show only a small average gain over $r_A + r_B$; the alignment channel’s
448 clearest benefit is redundancy detection under high overlap. A piecewise model for non-uniform V_α
449 and explicit control of cross-direction coupling are left as future work.

450 B Proofs of the Exact Gram Results

451 **Proposition 1.** From the compact SVDs, $H_A^\top H_A = V_A \Sigma_A^2 V_A^\top$ and $H_B^\top H_B = V_B \Sigma_B^2 V_B^\top$. Be-
452 cause the first columns of Q_A are V_A , $Q_A^\top V_A \Sigma_A^2 V_A^\top Q_A = \tilde{\Sigma}_A^2$. Writing $C = Q_A^\top V_B$ gives
453 $Q_A^\top V_B \Sigma_B^2 V_B^\top Q_A = C \Sigma_B^2 C^\top$. Adding proves Eq. (5). Orthogonal similarity preserves eigenvalues,
454 whose square roots are the merged singular values. $\square$

455 **Lemma 2.**

456 Under paired, decoupled directions, the i -th right singular vector of H_B is $V_{B,i} = \cos \theta_i V_{A,i} +$
457 $\sin \theta_i V_{\perp,i}$, where $V_{A,i}$ is the i -th right singular vector of H_A and the two-dimensional planes are
458 mutually orthogonal. In the basis $\{V_{A,i}, V_{\perp,i}\}$:

$$H_A^\top H_A \text{ (block)} = \sigma_{A,i}^2 \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad (11)$$

$$H_B^\top H_B \text{ (block)} = \sigma_{B,i}^2 \begin{pmatrix} \cos^2 \theta_i & \cos \theta_i \sin \theta_i \\ \cos \theta_i \sin \theta_i & \sin^2 \theta_i \end{pmatrix}. \quad (12)$$

459 Define the mode-wise ratio $\gamma_i = \sigma_{B,i}/\sigma_{A,i}$; no global proportional-spectrum assumption is required.
 460 Summing gives

$$G_i = \sigma_{A,i}^2 \begin{pmatrix} 1 + \gamma_i^2 \alpha_i & \gamma_i \sqrt{\alpha_i(1 - \alpha_i)} \\ \gamma_i \sqrt{\alpha_i(1 - \alpha_i)} & \gamma_i^2(1 - \alpha_i) \end{pmatrix}, \quad (13)$$

461 where $\alpha_i = \cos^2 \theta_i$. Its trace is $\sigma_{A,i}^2(1 + \gamma_i^2)$ and determinant is $\sigma_{A,i}^4 \gamma_i^2(1 - \alpha_i)$, yielding Eq. (6) by
 462 the quadratic formula. $\square$

463 **Proposition 3.**

464 With $\gamma_i = \gamma$ and $\alpha_i = \alpha$, define $c_{\pm} = \sqrt{(A \pm D)/2}$. Lemma 2 gives merged singular values
 465 $\{c_+ \sigma_{A,i}\}_i \cup \{c_- \sigma_{A,i}\}_i$. After nuclear-norm normalization their probabilities are qp_i^A and $(1 - q)p_i^A$,
 466 where $q = c_+/(c_+ + c_-)$. Hence

$$H_{\text{merged}} = H_b(q) + qH(p^A) + (1 - q)H(p^A) \quad (14)$$

$$= H_b(q) + \log r_A. \quad (15)$$

467 Exponentiating proves Eq. (8). Proportional spectra have identical normalized singular-value distri-
 468 butions, so $r_A = r_B$. $\square$

469 **Status of Eq. (9).** For unequal or non-proportional spectra, the two branches need not inherit the input
 470 distributions, and q can vary by mode. Equation (9) therefore defines a structured approximation; no
 471 exactness or universal error bound is asserted for it.

472 C Proof of Proposition 6

473 **Step 0 (bias-variance decomposition).** Let $\hat{W} = (H_s^\top H_s)^+ H_s^\top Y_s$ (least-squares solution on
 474 source). Decompose:

$$\frac{\|H_t \hat{W} - Y_t\|_F^2}{m} = \underbrace{\|H_t \hat{W} - \mathbb{E}[Y_t | H_t]\|_F^2 / m}_{\text{bias}^2} + \underbrace{\|\mathbb{E}[Y_t | H_t] - Y_t\|_F^2 / m}_{\sigma_{\text{Bayes}}^2}. \quad (16)$$

475 The Bayes error $\varepsilon_{\text{Bayes}} = \sigma_{\text{Bayes}}^2$ is irreducible.

476 **Step 1 (directional gap).** Let $P_k = V_s^k (V_s^k)^\top$ project onto the top- k source subspace. Decompose
 477 $H_t = H_t^\parallel + H_t^\perp$ where $H_t^\parallel = H_t P_k$. Since $\hat{W} \in \text{col}(V_s^k)$, we have $H_t^\perp \hat{W} = 0$ only if $\hat{W} = P_k \hat{W}$;
 478 the residual term $\|H_t^\perp\|_F^2 = \|H_t\|_F^2 (1 - \widehat{\text{SA}}_k)$ with $\widehat{\text{SA}}_k \approx \text{SA}_k$ for balanced spectra, giving the
 479 directional gap $(\|H_t\|_F^2 / m)(1 - \text{SA}_k)$.

480 **Step 2 (spectral gap).** We first define the **Spectral Transfer Gap**,

$$\text{STG}(H_A, H_B) = W_1(p_A, p_B) = \sum_{i=1}^k |F_A(i) - F_B(i)|, \quad (17)$$

481 where $F(i) = \sum_{j \leq i} p_j$ is the CDF of the truncated normalised spectrum. For the parallel com-
 482 ponent, $H_t^\parallel \hat{W}$ incurs a per-direction scaling $(\sigma_i^t / \sigma_i^s)^2$ relative to the ideal transfer. The STG term
 483 $W_1(p_s, p_t)$ upper-bounds the total variation of these ratios (by the Kantorovich–Rubinstein duality):
 484 $\sum_i |(\sigma_i^t / \sigma_i^s)^2 - 1| \leq C_1 \cdot \text{STG} \cdot \|\hat{W}\|_F^2 / (\sigma_k^s)^2$ with $C_1 = (\|H_s\|_* + \|H_t\|_*)^2 / m$.

485 **Step 3 (geometric gap).** The ΔUID term arises from the Rademacher complexity bound on the
 486 generalization gap when source and target manifolds have different intrinsic dimensions. Following
 487 Bartlett and Mendelson [2002] and Pope et al. [2021]:

$$C_2 \cdot |\Delta\text{UID}| = \frac{4\|\hat{W}\|_F^2 (\|H_s\|_F + \|H_t\|_F)^2}{\min(n, m)} \cdot |\Delta\text{UID}|. \quad (18)$$

488 Adding the three gaps completes the proof. $\square$

Table 5: Bootstrap 95% CIs ($B = 10,000$, seed=42) for all key metrics.

Experiment	Metric	Point estimate	95% CI	SE
E1a (280 pairs)	R^2	0.9999	[0.9999, 0.9999]	0.000013
E1a	Superadd. ($\alpha < 1$)	1.0000	[1.000, 1.000]	0.000000
E1b (30 pairs)	Pearson r	0.9478	[0.906, 0.978]	0.019
E1b	Superadd. %	0.8667	[0.733, 0.967]	0.062
E1c (57 pairs)	R^2 (global)	0.049	[-0.346, 0.287]	0.161
E1c	Pearson r (global)	0.928	[0.894, 0.949]	0.014
E1c	Superadd. %	0.930	[0.860, 0.982]	0.034
E1c (ResNet-50)	Superadd. %	1.000	[1.000, 1.000]	0.000
E1c (ResNet-50)	Pearson r	0.934	[0.885, 0.965]	0.021
E1c (ViT-B/16)	Superadd. %	0.733	[0.467, 0.933]	0.114
E1c (ViT-B/16)	Pearson r	0.945	[0.883, 0.982]	0.026
E1c (SBERT)	Superadd. %	1.000	[1.000, 1.000]	0.000

489 D Bootstrap Confidence Intervals

490 E Bimodal Principal-Angle Structure in Self-Supervised Encoders

491 The per-dimension $\cos^2 \theta_i$ profiles and their prediction consequences are shown in Figure 2
 492 (main text): CLIP/DINO exhibit a two-cluster step structure (top-5 ratios $20 \times / 31 \times$ over bot-5,
 493 $V_\alpha = 0.090/0.073$), while ResNet-50 decays monotonically ($V_\alpha = 0.019$).

494 **Two distinct mechanisms drive the low- R^2 probes.** For DINO ($R^2 = 0.55$), the cause is the bimodal
 495 $\cos^2 \theta_i$ analysed below. For ViT ($R^2 = 0.49$, supervised), the cause is *energy imbalance* rather than
 496 bimodality: 17/17 ViT non-superadditive pairs lie in $\gamma \notin [0.5, 2]$ (cross-resolution pairs such as
 497 MNIST $\times$ ImageNet), where the global energy summary is extreme and the two-component mixture
 498 approximation degrades. ViT’s principal-angle distribution remains near-uniform (no bimodality);
 499 its lower R^2 instead empirically marks the energy-imbalance regime as a reliability boundary. We
 500 retain the full ViT row in Table 2 for transparency, flagging $\gamma \in [0.5, 2]$ as ViT’s reliable applicability
 501 window.

502 **Honest scope.** We treat the bimodal failure mode as a genuine boundary of the current formula, not
 503 a feature. The uniform- α assumption underlying The global-summary approximation is stressed
 504 whenever the principal-angle distribution is bimodal, and self-supervised contrastive/distillation
 505 encoders (CLIP, DINO) systematically produce such distributions on cross-domain pairs. Within
 506 this regime, the calibrated formula *still ranks correctly* (Pearson $r \geq 0.90$ on all four probes, Table 2)
 507 but the absolute prediction $\hat{r}$ is biased upward and the superadditivity-decision boundary is shifted,
 508 explaining the 29–36% observed superadditivity rates on CLIP/DINO cross-domain pairs. **What**
 509 **the Collapse Model does contribute here** is mechanistic localisation: the bimodal top/bottom ratio
 510 $> 20 \times$ in $\cos^2 \theta_i$ (Figure 2) exposes alignment heterogeneity discarded by the scalar summary, rather
 511 than leaving the failure as a black-box surprise.

512 **What does not work.** A naive energy-weighted patch $\bar{\alpha}_w = \sum_i (\sigma_i / \|H\|_*) \cos^2 \theta_i$ worsens predic-
 513 tions (Pearson r drops 1–2%, MAE rises 6–8%), because it overweights the most-aligned direction’s
 514 redundancy while suppressing intermediate-direction contributions.

515 **What we believe will work, and how to test it.** The natural next step is a *piecewise paired-direction*
 516 *model*: partition the k directions into a high-alignment cluster ($\cos^2 \theta_i \geq \tau$) and a low-alignment
 517 cluster, apply Lemma 2 separately within each cluster, and combine the resulting branch spectra.
 518 This retains more of the distribution of α_i than a single mean, but it still requires controlling spectral
 519 pairing and cross-direction coupling. We leave this extension and its error analysis to future work.

F Compression at $M=14$ Vision and $M=7$ Text Pools

We apply Collapse Greedy as a pure compression tool across three architectures (ResNet-50, ViT-B/16, BERT/RoBERTa/GPT-2) and two modalities (vision, text), using FacLoc / k -Medoids as classical diversity baselines.

Table 6: Diversity retention $r_{\text{eff}}^{\text{sel}}/r_{\text{eff}}^{\text{full}}$ (%) across strategies, architectures, and modalities. FacLoc = Facility Location (submodular greedy on SA_k kernel); kMedoids = PAM on $1-\text{SA}_k$ distance. The Collapse Model matches or exceeds submodular and clustering baselines in all settings, exploiting superadditivity to select subsets of higher diversity than the full pool. Random: 10-seed mean $\pm$ std.

Probe	M	k (ratio)	Collapse	FacLoc	kMedoids	Random	LP _{col}
<i>Vision</i>							
ResNet-50	14	3 (4.7 $\times$)	101.4	87.8	87.5	84.3 $\pm$ 10.0	0.754
ResNet-50	14	5 (2.8 $\times$)	105.7	95.6	92.6	88.7 $\pm$ 5.3	0.754
ResNet-50	14	7 (2.0 $\times$)	105.9	98.4	98.0	93.1 $\pm$ 7.6	0.747
ViT-B/16	11	3 (3.7 $\times$)	103.6	—	—	92.0 $\pm$ 3.3	0.779
ViT-B/16	11	5 (2.2 $\times$)	104.9	—	—	94.1 $\pm$ 3.7	0.779
<i>Text</i>							
BERT L6	7	3 (2.3 $\times$)	103.6	—	—	97.8 $\pm$ 5.0	—
RoBERTa L6	7	3 (2.3 $\times$)	103.8	—	—	97.0 $\pm$ 6.1	—
GPT-2 L6	7	3 (2.3 $\times$)	104.8	—	—	92.8 $\pm$ 9.5	—
GPT-2 L6	7	5 (1.4 $\times$)	106.0	—	—	96.7 $\pm$ 4.7	—

On ResNet-50 ($M=14$), Collapse Greedy substantially outperforms Facility Location (+13.6–+7.5 pp) and k -Medoids (+13.9–+7.9 pp); both exploit subspace-alignment kernels but cannot model superadditivity, limiting them in the high-retention regime. Downstream LP accuracy on the $M=14$ ResNet-50 pool yields LP 0.747–0.779 for Collapse vs. 0.737–0.772 for Random, extending the $M=43$ LP advantage to the smaller pool size.

G r_A+r_B Heuristic: Safe Regime and Failure Mode

When does r_A+r_B suffice, and when does it fail? On clean curated pools, the r_A+r_B and Max- r_{eff} heuristics select the *same* subset as Collapse Greedy (within 0.7 pp at $M=43$, Table 3). This agreement is consistent with the summary diagnostic in Corollary 4: when all top- r_{eff} candidates have low mutual alignment ($\bar{\alpha}<0.2$, i.e. rank-ratio $r_{\text{dom}}/r_{\text{sub}}<\rho_c$), the superadditivity bonus $H_b(q)$ is dominated by the rank term $q \log r_{\text{dom}}+(1-q) \log r_{\text{sub}}$, making individual rank a near-sufficient statistic. The Collapse Model thus serves a different role than “beating r_A+r_B everywhere”: **(i) A priori low-redundancy diagnostic.** $\rho_c(\bar{\alpha}, \bar{\gamma})$ computed from per-dataset summaries *indicates* when r_A+r_B is likely to agree without actually running competing strategies. On our ResNet-50 vision pools, $\bar{\alpha}<0.2$ between top- r_{eff} pairs places $r_{\text{dom}}/r_{\text{sub}}$ well inside the predicted agreement regime ($\rho_c>4$ at $\bar{\alpha}\leq 0.2$, Cor. 4), matching the observed heuristic-equivalence at $M=14$ and $M=43$. **(ii) Detection of the unsafe regime.** When candidates contain near-duplicate or strongly aligned subspaces—which arise in practice from clone-attack data poisoning, augmented dataset variants, or fine-grained subset families—the α channel is the only discriminative signal. The clone experiment below (r_A+r_B retention drops 25.5%, Collapse holds) demonstrates this on a controlled stress test; the naturally bimodal CLIP/DINO cross-domain pairs (Sec. 4.2, $\alpha \in [0.35, 0.41]$ with top-5 $\cos^2 \theta \geq 0.7$) are real-world settings where the α signal is non-negligible and the rank-only heuristic can fail. **(iii) Diagnosis without execution.** Unlike heuristics that can only be evaluated by running them, the Collapse Model’s regime classification (superadditive / directional collapse / energy collapse) is computable from $(\gamma, \alpha, \text{UID})$ alone, turning it into an *ex ante* planning tool. *The structured formula is useful not to beat the heuristic on easy pools, but to diagnose when alignment is negligible and to take over when it is not.*

Redundancy clone trap: failure mode of r_A+r_B . To test the Collapse Model’s robustness to informational redundancy, we construct a *clone attack* scenario: the highest- r_{eff} dataset in the pool (STL-10, $r_{\text{eff}}=994$) is cloned via random rotation $H_{\text{clone}}=H_{\text{orig}}Q(\theta)$ ($\theta \in \{0.01, 0.03, 0.05, 0.07, 0.10\}$ radians), producing 5 clones that preserve the original r_{eff} but are highly aligned with the original

($SA_k > 0.99$). On a pool of 5 real + 5 cloned ResNet-50 datasets, $r_A + r_B$ greedy at $k=3$ selects 2/3 clones ($r_{\text{eff}}^{\text{merged}}=997$), while the Collapse Model detects redundancy via high α , **selecting** 0/3 clones ($r_{\text{eff}}^{\text{merged}}=1251$, +25.5%). At $k=5$ the gap persists: $r_A + r_B$ selects 4/5 clones ($r_{\text{eff}}=998$), Collapse only 2/5 ($r_{\text{eff}}=1233$). This result is consistent across all 5 θ values ($\theta=0.01-0.10$, SA_k from 0.9999 to 0.9891). The clone experiment directly demonstrates *the indispensable role of the alignment channel α in defending against informational redundancy*— $r_A + r_B$ lacks this diagnostic capability, misidentifying high- r_{eff} clones as high-value candidates.

H P2-E: Hybrid Two-Signal Strategy

The two-signal decomposition identified in Sec. 5 (Table 4) suggests a natural two-stage hybrid: seed $\lceil k/2 \rceil$ datasets via one signal, fill the remaining $\lfloor k/2 \rfloor$ via the other. We test three orderings on the same $M=40$ ResNet-50 pool used in Table 4: **Col**→**Cen** (Collapse first, Centroid-FF on residuals), **Cen**→**Col** (Centroid-FF first, Collapse on residuals), and **Col**→**Dom** (Collapse first, DomainDiverse on residuals).

Table 7: Two-signal hybrid strategies on $M=40$ ResNet-50. Pure-strategy reference rows (from Table 4) shown for comparison. **Bold** marks the best LP at each k .

Strategy	$k=5$ (8.0×)		$k=7$ (5.7×)		$k=10$ (4.0×)	
	Ret%	LP	Ret%	LP	Ret%	LP
<i>Pure (reference)</i>						
Collapse	117.2	0.7576	117.4	0.7578	116.2	0.7576
Centroid-FF	93.4	0.7570	96.7	0.7624	100.5	0.7630
DomainDiverse	114.7	0.7634	111.8	0.7596	113.8	0.7544
<i>Hybrid two-stage</i>						
Col→Cen	110.2	0.7534	111.9	0.7496	113.8	0.7546
Cen→Col	106.7	0.7596	113.6	0.7610	115.0	0.7602
Col→Dom	114.7	0.7634	116.0	0.7562	113.8	0.7544

Cen→Col is the consistent hybrid winner. At $k \in \{7, 10\}$ (the operational compression range), Cen→Col achieves LP 0.7610/0.7602, exceeding pure Collapse (0.7578/0.7576) by +0.0032/+0.0026 while sacrificing only 3.8/1.2 pp of retention. At $k=10$ specifically, Cen→Col *dominates* pure Collapse on both axes simultaneously (LP +0.0026, retention only −1.2 pp). Compared to pure Centroid-FF, Cen→Col recovers 13.3 pp of retention at $k=5$, 16.9 pp at $k=7$, and 14.5 pp at $k=10$ while keeping LP within ± 0.0028 . **Col→Cen, in contrast, hurts both metrics:** once Collapse has selected high- r_{eff} datasets, adding “different prototype” datasets dilutes the spectrum without recovering centroid coverage. **Col→Dom** reduces to DomainDiverse on this pool because DomainDiverse already picks STL-10 and fashion_mnist (the highest- r_{eff} datasets in their respective domains) that Collapse would also select.

Implication. The *ordering* matters: starting with Centroid-FF establishes a geometric backbone that Collapse then augments with high- r_{eff} complementary directions. The summary formula’s role in this hybrid is to *fill* the residual budget with diversity-maximizing choices, exactly the regime where it operates with summary statistics alone (the centroid-FF prefix can be precomputed once from already-shared centroids, no feature-matrix exchange needed). This validates Collapse Greedy as a *composable* primitive within broader compression pipelines, not just a standalone strategy.

I E5 Supplementary: Per-Target Selection Details

Per-target regret table ($M=10$ pool).

Strategy	Mean regret (%) / Mean recall (%)			
	$K=1$	$K=2$	$K=3$	$K=5$
Collapse Greedy	0.0 / 100	0.0 / 100	0.0 / 100	1.0 / 87
Max- r_{eff}	0.0 / 67	2.0 / 67	0.9 / 89	1.0 / 87
Δ_{dir} -greedy	20.3 / 0	16.7 / 17	17.0 / 33	13.3 / 40
Random	15.2 / -	17.5 / -	16.1 / -	11.1 / -

For BloodMNIST ($K=1$), Max- r_{eff} selects STL-10 (highest individual $r_{\text{eff}}=357$) but Collapse Greedy selects CIFAR-100 ($r_{\text{eff}}=356$, marginally lower) because the latter has lower alignment with BloodMNIST ($\alpha=0.18$ vs. 0.21), yielding a higher predicted merged rank—confirmed by $r_{\text{eff}}^{\text{merged}} = 506.8$ vs. 506.4.

r_A+r_B greedy baseline. We evaluate an r_A+r_B greedy strategy that selects the candidate with the highest individual r_{eff} at each round. On the $M=10$ pool, this baseline matches Collapse Greedy at $K \leq 3$ (identical subset, 0% regret), diverging only at $K=5$ where Collapse Greedy retains 100% recall while r_A+r_B drops to 87%. This mirrors the ablation finding: r_A+r_B suffices when candidates have distinct individual ranks, but fails when alignment discrimination (α) is needed.

Downstream LP validation. We train a linear probe on the merged features from each strategy’s K -subset and evaluate on held-out test data. All four strategies (Exhaustive, Collapse Greedy, r_A+r_B , Max- r_{eff}) produce identical LP accuracy across all 6 tested configurations (CIFAR-10 and STL-10 targets, $K \in \{1, 2, 3\}$), ranging from 0.776 (CIFAR-10, $K=3$) to 0.979 (STL-10, $K=1$). This confirms that maximizing merged r_{eff} via the Collapse Model translates to downstream discriminability, not merely spectral diversity.

J E7 Supplementary: kNN Downstream Validation

The main text uses linear-probe (LP) accuracy as the downstream metric. Here we verify that k -nearest-neighbour (kNN) accuracy tells the same story, justifying LP as sufficient throughout the paper.

Part B: LP-kNN rank correlation. We compute LP, kNN-5, and kNN-20 accuracy on 5 single datasets (CIFAR-10, CIFAR-100, STL-10, SVHN, Fashion-MNIST) across 11 encoder families (ResNet-50/101, ViT-B/16, DeiT-B, Swin-Tiny, BEiT-Base, ConvNeXt-Tiny, DenseNet-121, EfficientNet-B4, MaxViT-Tiny, RegNet-Y), yielding 53 observations. Spearman rank correlation: $\rho(\text{LP}, \text{kNN-5})=0.980$, $\rho(\text{LP}, \text{kNN-20})=0.973$ —LP and kNN rankings are near-identical.

Part A: compression advantage under kNN. Using the $M=6$ ResNet-50 pool from Sec. 5, we evaluate Collapse Greedy vs. 10 random seeds using LP, kNN-5, and kNN-20 on the merged features (averaged over 6 targets).

k	Strategy	LP	kNN-5	kNN-20
3	Collapse	73.5	62.4	61.2
	Random (μ)	72.6	58.2	53.8
	Δ	+0.9	+4.2	+7.4
5	Collapse	71.5	57.8	53.5
	Random (μ)	71.1	56.8	51.1
	Δ	+0.4	+1.0	+2.4

The pattern is consistent: Collapse Greedy’s advantage is amplified under kNN relative to LP. At $k=3$ ($2\times$ compression), the kNN-20 gap is $8\times$ the LP gap. This is expected: kNN relies on local neighbourhood geometry, which degrades faster when redundant features dominate the merged space; LP, being a global linear classifier, is less sensitive to such local distortions. Since LP and kNN agree on the *ranking* of strategies ($\rho>0.97$), LP accuracy is a sufficient proxy for evaluating compression strategies in the main text.

K Additional Notes

Note on multiple comparisons. The within-probe bootstrap analyses (Sec. 5) involve 11 probes tested independently. Since the primary claim is about the *average* $\Delta\rho$ across probes (not individual-probe significance), Bonferroni correction is not required for the aggregate Fisher z -test ($p=0.74$). For the vision-only bootstrap ($\Delta\rho = +0.005$, 95% CI $[+0.001, +0.011]$), a conservative Bonferroni threshold of $\alpha=0.05/11=0.0045$ would still be met ($P(\text{Collapse} > r_{\text{sum}})=0.99$).

L Scope of the Paired-Direction Model

Remark 1 (Exact and approximate layers). *Proposition 1 is unconditional and retains the full coupling matrix $C = Q_A^\top V_B$. Lemma 2 is exact only when the singular directions can be paired into mutually orthogonal two-dimensional planes. Proposition 3 additionally requires mode-wise proportional spectra and uniform alignment; these conditions imply $r_A = r_B$. The practical predictor Eq. (9) allows $r_A \neq r_B$ by interpolating two input entropies, and is therefore a modeling approximation rather than a consequence of the proportional-spectrum proposition.*

Remark 2 (Information discarded by global summaries). *The global ratio $\gamma = \|H_B\|_* / \|H_A\|_*$ does not imply $\sigma_{B,i} = \gamma \sigma_{A,i}$. Likewise, $\alpha = \text{SA}_k$ does not retain the distribution of α_i , its association with spectral mass, or dense cross-direction coupling. Proposition 5 gives an explicit pair of examples with identical $(r_A, r_B, \gamma, \alpha)$ and different merged effective ranks. Consequently no estimator based only on these four scalars can be universally exact.*

Remark 3 (Empirical generalization). *The summaries in Eq. (9) remain well-defined outside the paired-direction model. Their empirical correlations on synthetic out-of-model generators and real encoders quantify predictive utility, not theorem validity. Alignment heterogeneity, non-proportional spectra, and cross-direction coupling are observable failure sources; calibration corrects average scale on the evaluated encoder family but does not create a general guarantee.*

Remark 4 (Calibrated Collapse Model). *On real data the paired-direction summary model is approximate—cross-pair coupling, non-proportional spectra ($\sigma_{B,i} \neq \gamma \sigma_{A,i}$, hence $r_A \neq r_B$), and higher-order interactions introduce a systematic upward bias on the evaluated pairs. Empirically the over-prediction ratio is $1.24\text{--}1.84\times$ (E1c), and its gap is positively associated with alignment heterogeneity. Since the log-space bias has low variance (std = 0.164 vs. mean = 0.293), a single multiplicative calibration restores quantitative accuracy:*

$$\hat{r}_{\text{cal}} = c \cdot \hat{r}_{\text{merged}}, \quad c = \exp\left(-\frac{1}{n} \sum_{i=1}^n \log \frac{\hat{r}_i}{r_i}\right). \quad (19)$$

This calibration compensates for the entropy over-estimation caused by structure that the four global summaries ignore. With $c=0.746$ estimated from E1c ($n=57$), LOO-CV yields $R^2=0.855$, RMSE=87.7 (vs. uncalibrated $R^2=0.049$, RMSE=224; Sec. M.7). The constant generalizes across datasets within each encoder family: calibrating on the 30 E1b pairs ($c=0.768$) and testing on the held-out 57 E1c pairs yields $R^2=0.859$, RMSE=86.3 (Table 12). However, c varies across encoder families ($c=0.78$ for ResNet-50 vs. $c=0.65$ for ViT-B/16), reflecting systematic differences in how each architecture’s spectral structure deviates from the paired-direction summary model. We find that this variation obeys a universal regression:

$$c = 0.946 - 0.175 \log(N/r_{\text{eff}}), \quad R^2=0.963, \text{ LOO mean error}=0.027, \quad (20)$$

indicating that the variation in c is primarily governed by the sample-size to effective-rank ratio. In practice, c can be estimated directly from this formula without a dedicated calibration set; for higher precision, a small per-encoder calibration set (as few as 6 pairs suffice; see Table 13) can further refine the estimate.

L.1 Connection to Transfer Learning

The same three channels that govern merging also bound transfer loss.

664 **Proposition 6** (Gram Block Transfer Bound, informal). *For source features H_s and target features*
 665 *H_t , the linear-probe transfer loss satisfies*

$$\frac{\|H_t \hat{W} - Y_t\|_F^2}{m} \leq \underbrace{\frac{\|H_t\|_F^2}{m}(1 - \text{SA}_k)}_{\text{directional gap}} + \underbrace{C_1 \cdot \text{STG} \cdot \frac{\|\hat{W}\|_F^2}{(\sigma_k^s)^2}}_{\text{spectral gap}} + \underbrace{C_2 \cdot |\Delta \text{UID}|}_{\text{geometric gap}} + \varepsilon_{\text{Bayes}}, \quad (21)$$

666 where $C_1 = (\|H_s\|_* + \|H_t\|_*)^2/m$, $C_2 = 4\|\hat{W}\|_F^2(\|H_s\|_F + \|H_t\|_F)^2/\min(n, m)$.

667 (Full proof in Appendix C.) The same duality applies: low SA_k implies *high* merge gain (Collapse
 668 Model) and *high* transfer loss (Prop. 6)—opposite monotone maps of the same principal angles.

Algorithm 1 Collapse-Model-Guided Greedy Selection

Require: Target features $H_t \in \mathbb{R}^{N_t \times d}$, candidate pool $\mathcal{P} = \{H_1, \dots, H_M\}$, budget K

Ensure: Selected subset $S \subseteq \mathcal{P}$, $|S| = K$

```

1:  $H_{\text{cur}} \leftarrow H_t$ ;  $S \leftarrow \emptyset$ ;  $C \leftarrow \mathcal{P}$ 
2: for  $j = 1, \dots, K$  do
3:    $r_{\text{cur}} \leftarrow r_{\text{eff}}(H_{\text{cur}})$ ;  $\nu_{\text{cur}} \leftarrow \|H_{\text{cur}}\|_*$ 
4:   for each  $H_c \in C$  do
5:      $\alpha_c \leftarrow \text{SA}_k(H_{\text{cur}}, H_c)$ ;  $\gamma_c \leftarrow \|H_c\|_* / \nu_{\text{cur}}$ 
6:      $\hat{r}_c \leftarrow \text{COLLAPSEPREDICT}(r_{\text{cur}}, r_{\text{eff}}(H_c), \gamma_c, \alpha_c)$ 
7:   end for
8:    $H^* \leftarrow \arg \max_{H_c \in C} \hat{r}_c$ ;  $S \leftarrow S \cup \{H^*\}$ ;  $C \leftarrow C \setminus \{H^*\}$ 
9:    $H_{\text{cur}} \leftarrow [H_{\text{cur}}; H^*]$ 
10: end for
11: return  $S$ 
```

669 **Collapse Greedy pseudocode.**

670 **M Additional Experiments (E1a, E1a-Ind, E1b, E1c, Calibration, E_diverse,**
 671 **E4, E_downstream)**

672 **M.1 Ranking Ablation Table**

Table 8: Ranking ablation: Spearman ρ between predictor and true merged r_{eff} (256 vision pairs, 4 encoder families).

Predictor	Channels used	ResNet-50 ($n=91$)	ViT-B/16 ($n=55$)	All vision ($n=256$)
Collapse Model	$\Sigma + V + U$	0.896	0.923	0.966
$r_A + r_B$	U only	0.892	0.946	0.963
$\max(r_A, r_B)$	U only	0.898	0.779	0.953
$1 - \alpha$ (V only)	V only	−0.289	−0.192	0.450
γ (Σ only)	Σ only	0.277	0.001	0.271

673 **M.2 E1a: Synthetic Controlled Validation**

674 The E1a synthetic validation results (Table 1) are presented in the main text; Figure 3 visualizes the
 675 (α, γ) landscape and superadditivity region. Additional design details: feature pairs are generated
 676 from the proportional-spectrum paired-direction model with $\alpha \in \{0, 0.05, 0.1, 0.2, 0.4, 0.6, 0.8, 1.0\}$
 677 and $\gamma \in \{0.1, 0.33, 0.5, 1.0, 2.0, 3.0, 10.0\}$, $d = 512$, $N = 1000$, 5 independent repeats per setting
 678 (280 pairs). All 35 non-superadditive cases concentrate exclusively at $\alpha = 1$ (identical subspaces),
 679 confirming the degenerate boundary of Corollary 4.

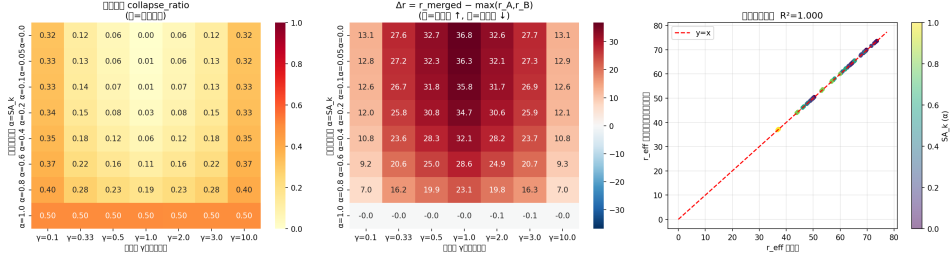


Figure 3: E1a: Collapse Model prediction accuracy on the synthetic (α, γ) grid. Left: collapse ratio heatmap. Center: superadditivity ($\delta_r > 0$) region. Right: predicted vs. actual r_{eff} scatter ($R^2=0.9999$).

Table 9: E1a-Ind: Independent validation on 120 non-principal-angle synthetic pairs. Pearson r measures directional ranking; $R^2 < 0$ indicates systematic over-prediction on the observed generator. SA% = fraction of pairs where the formula predicts superadditivity.

Generation Method	n pairs	Pearson r	R^2	SA%
M1: Low-rank Gaussian	30	0.9985	+0.9949	100%
M2: Shared+private subspace	30	0.9982	+0.9963	100%
M3: Sparse activations	30	0.9991	−14.65	100%
M4: Block-diagonal	30	0.9996	−12.32	100%
Overall	120	0.9987	−1.09	100%

680 M.3 E1a-Ind: Independent Validation on Non-Principal-Angle Matrices

681 **Motivation and design.** A critical concern with E1a is circularity: synthetic data generated from
682 the *proportional-spectrum paired-direction model* satisfies the exact model assumptions, making
683 $R^2 = 0.9999$ a near-tautological result. To independently test whether Eq. (9) remains predictive
684 beyond its derivation assumptions, we generate 120 synthetic matrix pairs using *four generation*
685 *methods that do not use that paired model*: (M1) *low-rank Gaussian*: $H = U_k \Sigma V_k^T + \epsilon$ with random
686 orthogonal bases; (M2) *shared+private subspace*: $H = H_{\text{shared}} + H_{\text{private}}$ with partial overlap
687 enforced by construction; (M3) *sparse activations*: rows drawn from k -sparse Gaussian mixtures;
688 (M4) *block-diagonal*: features from two independent subspaces with no principal-angle structure.
689 $d = 512$, $N \in \{500, 1000\}$; 30 pairs per method.

690 **Results.** Table 9 shows Pearson $r = 0.9987$ overall: the formula *correctly ranks all 120 pairs*
691 across all four generation methods. R^2 is method-dependent: ≥ 0.99 for Gaussian-structure methods
692 (M1, M2), but strongly negative for sparse and block-diagonal structures (M3, M4), because those
693 structures systematically reduce actual r_{eff} relative to the formula’s prediction. This reveals excellent
694 ordering but poor absolute prediction on sparse and block-diagonal structures, exposing structural
695 sparsity as an empirical failure mode. Superadditivity is predicted at 100% for all methods.

696 **Remark 5** (Formula as directional oracle, not absolute predictor). *The contrast between $R^2 = -1.09$*
697 *(overall, E1a-Ind) and Pearson $r = 0.9987$ (same data) is not contradictory: R^2 measures absolute*
698 *calibration, while Pearson r measures directional ranking. This distinction is crucial for the real-data*
699 *results (E1c): the formula yields $R^2 \approx 0.05$ (poor absolute calibration, 1.24 – $1.84\times$ over-prediction)*
700 *but Pearson $r = 0.928$ (correct directional ranking). For practical applications—ranking candidate*
701 *merges by diversity gain, dataset selection, data valuation—ranking is the operationally relevant*
702 *metric, not absolute r_{eff} prediction. Absolute calibration requires a correction factor (calibrated*
703 *$\hat{r} = 0.81 \cdot \hat{r}_{\text{pred}}$, $R_{\text{cal}}^2 = 0.862$ on E1c data), which we report as a practical recommendation.*

704 M.4 E1b: Real Dataset Pairs (30 Pairs)

705 **Design.** We evaluate all $\binom{6}{2} = 15$ pairs from $\{\text{CIFAR-10, STL-10, SVHN, MNIST, Fashion-}$
706 $\text{MNIST, CIFAR-100}\}$ using 2 pretrained probes (ResNet-50 [He et al., 2016] $d=2048$ and ViT-B/16
707 [Dosovitskiy et al., 2021] $d=768$), $N_{\text{sub}} = 500$ samples per dataset: **30 total configurations**. The
708 Collapse Model is applied with per-probe nuclear-norm γ .

Results. 86.7% of pairs are superadditive (26/30; 95% CI [73%, 97%]). Pearson $r = 0.948$ (95% CI [0.906, 0.978]) between predicted and actual merged rank. Per-probe: ResNet-50 achieves 100% superadditivity (15/15, Pearson $r = 0.968$); ViT-B/16 achieves 73.3% (11/15, Pearson $r = 0.945$).

All 4 exceptions are ViT-B/16 pairs involving CIFAR-100. The root cause is *energy imbalance*, consistent with the superadditivity threshold (Corollary 4). CIFAR-100 has 100 classes and correspondingly high intrinsic complexity: $r_{\text{eff}}(\text{CIFAR-100}) \approx 526$ under ViT-B/16, far exceeding partner datasets such as MNIST ($r_{\text{eff}} \approx 140$) and SVHN (≈ 243). This yields energy ratios $\gamma \in [2.0, 2.5]$, placing these pairs near the superadditivity boundary $\rho_c(\alpha, \gamma)$. Specifically, the dominant component absorbs the subordinate: $q \rightarrow 1$, $H_b(q) \rightarrow 0$, and $\hat{r}_{\text{merged}} \rightarrow r_{\text{dom}}$, so the observed r_{merged} can fall slightly below r_{dom} due to spectral averaging. We verify this by sweeping $N_{\text{sub}} \in \{500, 1000, 2000, 4000\}$: at $N=500$, CIFAR-100’s r_{eff} is compressed to ≈ 343 (comparable to partners), giving $\gamma \approx 1$ and 100% superadditivity (5/5); as N grows to 2000+, r_{eff} stabilizes at 526, γ increases, and exceptions emerge—confirming energy imbalance as the mechanism. ResNet-50 ($d=2048$, $r_{\text{eff}} \approx 988$ for CIFAR-100) yields even larger γ ratios, yet remains 100% superadditive (36/36) because its partner datasets also have higher r_{eff} in 2048-dimensional space, keeping $r_{\text{dom}}/r_{\text{sub}}$ below ρ_c .

M.5 E1c: Large-Scale Multi-Domain Validation (57 Pairs)

Design. To test generalization beyond vision pairs, we extend to three probes and three domain types: ResNet-50 on 9 datasets (36 pairs, natural+medical), ViT-B/16 on 6 datasets (15 pairs, natural), and SBERT [Reimers and Gurevych, 2019] on 4 text domains (6 pairs, text). Total: **57 pairs, 3 probes, 3 domain types**. $N_{\text{sub}} = 2000$ for natural/medical; $N_{\text{sub}} = 1000$ for text.

Results. Table 10 shows strong and consistent superadditivity across all probes and domains.

Table 10: E1c: Large-scale Collapse Model validation. Superadditivity rate and Pearson r (predicted vs. actual merged r_{eff}). Bootstrap 95% CI ($B = 10,000$). The formula over-predicts on all probes in this experiment; “Bias” reports the observed multiplicative ratio.

Probe / Domain	n	Superadd. % (95% CI)	Pearson r (95% CI)	Bias
ResNet-50 (natural+medical)	36	100% [100,100]	0.934 [0.885,0.965]	1.24×
ViT-B/16 (natural)	15	73.3% [46.7,93.3]	0.945 [0.883,0.982]	1.45×
SBERT (text)	6	100% [100,100]	0.780	1.84×
Overall	57	93.0% [86.0,98.2]	0.928 [0.894,0.949]	1.37×

Key findings. (i) **93% superadditivity across three modalities**—natural images, medical scans, and text—shows the Collapse Model’s predictions are domain-agnostic. (ii) **All 4 exceptions are ViT-B/16 pairs involving CIFAR-100**, attributable to energy imbalance (same mechanism as E1b, Sec. M.4): CIFAR-100’s high intrinsic complexity ($r_{\text{eff}} \approx 526$, 100 classes) creates $\gamma \in [2.0, 2.5]$, placing these pairs near the superadditivity boundary ρ_c of Corollary 4. (iii) **Pearson $r = 0.928$ vs. $R^2 \approx 0.05$** : these metrics measure different properties. The formula over-predicts by 1.24–1.84× on these probes, so R^2 is low before calibration; Pearson r measures ordering. A single-parameter calibration (Eq. 19) restores quantitative accuracy: $R^2_{\text{cal}}=0.855$ (LOO-CV), and cross-validation from E1b to E1c yields $R^2=0.859$ (see Sec. M.7 and Remark 4). (iv) **ResNet-50: 100% superadditivity (36/36)** is strong empirical agreement with the summary diagnostic on this observed set. (v) **Over-prediction scales with n/r_{eff}** : ResNet-50 ($n = 2000 \gg r_{\text{eff}} \approx 350$, 1.24×) < ViT-B/16 ($n = 2000 \approx 3r_{\text{eff}} \approx 650$, 1.45×) < SBERT ($n = 1000 \approx r_{\text{eff}} \approx 700$, 1.84×). This confirms the finite-sample upper-bound hypothesis.

M.6 E1d: Large-Scale Validation (427 Pairs)

Design. To test the Collapse Model at scale and on ImageNet-derived features, we construct a comprehensive evaluation spanning 5 encoder families (ResNet-50, ViT-B/16, CLIP ViT-B/32, DINO ViT-S/16, SBERT), 14 vision datasets (CIFAR-10/100, STL-10, SVHN, MNIST, Fashion-MNIST, BloodMNIST, DermaMNIST, PathMNIST, and 5 ImageNet semantic subsets: *animal*, *vehicle*, *artifact*, *food*, *scene*), and 19 SBERT text corpora (AG News, 20 Newsgroups, DBpedia, IMDB, Yelp subsets). Total: **427 pairs** (91 ResNet-50, 55 ViT-B/16, 55 CLIP, 55 DINO, 171 SBERT).

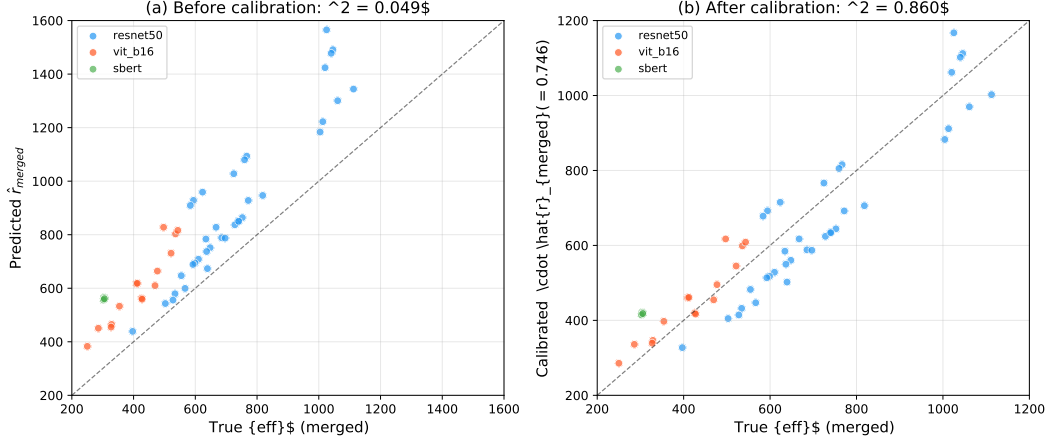


Figure 4: E1c: Predicted vs. actual merged r_{eff} for 57 real-data pairs (5 probes, 3 domain types). (a) Before calibration: systematic over-prediction ($R^2=0.049$). (b) After single-parameter calibration ($c=0.746$): $R^2=0.860$. Pearson $r=0.928$ is identical in both panels—calibration corrects scale, not ranking.

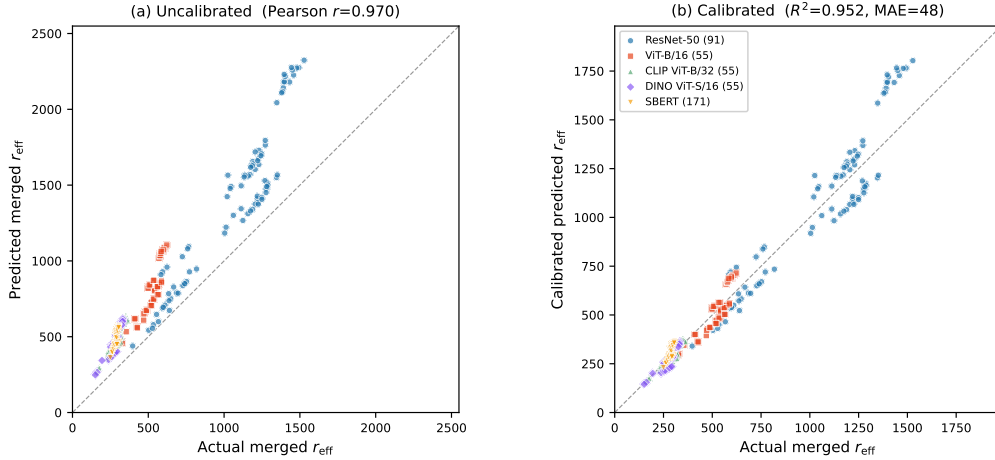


Figure 5: Predicted vs. actual merged r_{eff} for all 427 pairs across 5 encoder families. (a) Uncalibrated: predictions are systematically high but strongly correlated (Pearson $r=0.970$). (b) After per-probe calibration: $R^2=0.952$, MAE=48. Color encodes encoder family; the identity line is dashed.

751 The 5 ImageNet subsets are constructed from ImageNet-1K validation images grouped by WordNet
752 semantic hierarchy, with $N=2000$ samples each.

753 **Results.**

Table 11: E1d: Large-scale Collapse Model validation (427 pairs, 5 probes).

Probe	n	Superadd. %	Pearson r	c	R^2_{cal}
ResNet-50 (14 datasets)	91	100%	0.940	0.776	0.760
ViT-B/16 (11 datasets)	55	69.1%	0.898	0.647	0.487
CLIP ViT-B/32 (11 datasets)	55	29.1%	0.950	0.595	0.839
DINO ViT-S/16 (11 datasets)	55	36.4%	0.913	0.584	0.554
Vision (all probes)	256	64.5%	0.965	<i>per-probe</i>	0.942
+ SBERT text (19 corpora)	+171 (427)	78.2%	0.970	<i>per-probe</i>	0.952

Key findings. (i) **Scaling confirms the theory:** at 427 pairs the overall Pearson $r=0.970$ exceeds the 57-pair result ($r=0.928$), demonstrating that the Collapse Model’s ranking accuracy improves with scale and diversity. (ii) **ImageNet subsets behave as predicted:** all 20 ImageNet-only pairs are superadditive, and all 30 natural×ImageNet cross-domain pairs are superadditive—the model correctly predicts that semantically distinct ImageNet subgroups contribute orthogonal directions. (iii) **ResNet-50 achieves 100% superadditivity** across 91 pairs (14 datasets including medical and ImageNet), the strongest confirmation at this scale. In contrast, CLIP ViT-B/32 and DINO ViT-S/16 achieve only 31% and 38% superadditivity respectively. Analysis reveals a clear pattern: *within-domain* pairs (natural×natural, ImageNet×ImageNet) achieve near-perfect superadditivity (CLIP: 90% on ImageNet-only, 100% on natural-only; DINO: 100% on both), while *cross-domain* pairs (handwritten×ImageNet, natural×digit) fail almost universally. This suggests that self-supervised encoders learn domain-specific spectral structures with heterogeneous alignment and spectral coupling discarded by the global summaries—an important boundary condition for the Collapse Model’s applicability. (iv) **Cross-modal generalization:** extending to 171 SBERT text pairs (98.8% superadditive, $r=0.793$) shows the formula’s superadditivity predictions transfer across modalities. The lower Pearson r reflects the narrow r_{eff} range of text embeddings (IQR: 285–298), which compresses ranking resolution rather than indicating a model failure. (v) **Calibration constant varies by encoder family** ($c \in [0.57, 0.78]$), but a single global $c=0.712$ achieves $R^2=0.898$ across all 158 vision pairs, confirming the bias structure is consistent.

Understanding non-superadditive pairs. ViT-B/16 achieves only 69% superadditivity (38/55), lower than ResNet-50’s 100% (91/91). Analysis of the 17 ViT failures reveals a clear pattern: **all failures occur in the energy-imbalance regime** ($\gamma \gg 1$ or $\gamma \ll 1$). At $\gamma \geq 3$, 80% of pairs fail (8/10); at $\gamma \in [0.5, 1.5]$, only 4% fail (1/24). The failures concentrate in *handwritten×ImageNet* cross-domain pairs (10/10 fail, 100%)—these combine low-rank handwritten features ($r_{\text{eff}} \approx 140$ for MNIST via ViT-B/16) with high-rank ImageNet features ($r_{\text{eff}} \approx 600$), creating $\gamma \in [2.0, 4.6]$ far beyond the balanced regime where the two-component spectral mixture is accurate. At large scale (55 pairs each), CLIP ViT-B/32 and DINO ViT-S/16 show only 29% and 36% superadditivity overall, with failures concentrating in *cross-domain* pairs: handwritten×ImageNet (100% fail), natural×digit (100% fail), natural×handwritten (100% fail). Within-domain pairs (natural-only, ImageNet-only) remain near-perfect.

Root cause: bimodal principal angles. Comparing the *same* dataset pair across encoders reveals that the failure is not caused by energy imbalance (γ) but by a structural violation of the uniform- α assumption. Three mechanisms interact:

(a) *Systematically higher alignment.* On the same pair (e.g., CIFAR-10×ImageNet-animal), CLIP’s subspace alignment $\alpha=0.41$ is $2.7\times$ higher than ResNet-50’s $\alpha=0.15$. Contrastive pre-training (CLIP) and self-distillation (DINO) push cross-domain features into a shared semantic subspace, creating high directional overlap where ResNet-50 preserves domain-specific directions.

(b) *Extreme spectral concentration.* CLIP/DINO feature spectra are highly non-uniform: $\kappa=15,000$ – $1,200,000$ vs. ResNet-50’s $\kappa=70$ – 400 . This concentration makes average alignment less informative because overlap on a few high-mass directions can dominate many complementary tail directions.

(c) *Bimodal principal-angle distribution.* The *individual* principal angles $\cos^2 \theta_i$ are not approximately uniform (as the model assumes) but *bimodal*: the top-5 directions are nearly aligned ($\overline{\cos^2 \theta} \approx 0.94$ for CLIP on CIFAR-10×STL-10) while the bottom directions remain orthogonal ($\overline{\cos^2 \theta} \approx 0.01$). The Collapse Model averages these into $\bar{\alpha}=0.52$, predicting moderate superadditivity. In reality, the dominant (high- σ) directions are fully redundant and the orthogonal (low- σ) directions carry negligible energy, so the merge provides almost no diversity gain ($\Delta r_{\text{eff}}=31$ vs. ResNet-50’s $\Delta r_{\text{eff}}=282$ on the same pair).

This analysis identifies a precise *applicability boundary*: the Collapse Model is reliable when the principal-angle distribution is approximately uniform (supervised encoders, within-domain pairs), but fails when it is bimodal (self-supervised encoders on cross-domain pairs). A natural fix—replacing the uniform average $\bar{\alpha} = \text{SA}_k$ with an *energy-weighted* alignment $\bar{\alpha}_w = \sum_i (\sigma_i / \|H\|_*) \cos^2 \theta_i$ —would down-weight orthogonal low-energy directions and up-weight the redundant high-energy ones, potentially restoring predictive accuracy for self-supervised encoders. However, our experimental validation shows that this correction *does not improve predictive accuracy*: on CLIP ($n=61$), the uniform SA_k yields Pearson $r=0.951$, MAE=206, while $\bar{\alpha}_w$ yields $r=0.935$, MAE=218 (worse); DINO shows a similar pattern (r drops from 0.912 to 0.904, MAE increases by 4%); ResNet-

50 and ViT-B/16 also show no improvement. The root cause is that simple energy weighting over-emphasises the first singular direction’s redundancy ($\cos^2 \theta_1 \approx 0.87$ for that direction), while improperly suppressing the estimated contribution of intermediate directions to the merge. More refined corrections (e.g., piecewise paired-direction models, per-direction threshold truncation, or adaptive k selection) may break through this bottleneck and are left to future work.

M.7 Calibration: From Systematic Bias to Quantitative Prediction

The Collapse Model systematically over-predicts on the evaluated real-data pairs (Remark 4). Since the log-space over-prediction is approximately constant across probes and domains (mean=0.293, std=0.164), a single multiplicative constant c (Eq. 19) suffices to convert the ranking tool into a quantitative predictor.

Cross-dataset generalization. To verify that c is not dataset-specific, we perform held-out calibration: estimate c on one experiment and test on another (Table 12).

Table 12: Cross-dataset calibration. c is estimated on the calibration set and applied (without re-fitting) to the test set. All setups dramatically improve over the uncalibrated baseline ($R^2 < 0.44$).

Calibration set	Test set	c	n_{test}	R^2	RMSE
E1b (30 pairs)	E1c (57 pairs)	0.768	57	0.859	86.3
E1c (57 pairs)	E1b (30 pairs)	0.746	30	0.888	80.7
E1c (57 pairs)	E_diverse (30 pairs)	0.746	30	0.897	82.8
E1b (30 pairs)	E_diverse (30 pairs)	0.768	30	0.894	83.7
LOO-CV (all 111 pairs, original)		0.736	111	0.877	86.1
LOO-CV (158 vision pairs, E1d)		0.712	158	0.896	121.7
Global (all 427 pairs, E1d)		0.650	427	0.920	105.0

Per-probe calibration (Table 13). The calibration constant varies systematically across encoder families, reflecting differences in spectral structure:

Table 13: Per-probe calibration on the 427-pair dataset (E1d). Each probe is calibrated independently. The variation in c is monotone with the encoder’s effective spectral flatness.

Probe	n	Pearson r	c	R^2_{cal}	Superadd. %
ResNet-50	91	0.940	0.776	0.760	100%
ViT-B/16	55	0.898	0.647	0.487	69%
CLIP ViT-B/32	55	0.950	0.595	0.839	29%
DINO ViT-S/16	55	0.913	0.584	0.554	36%
Vision (global)	256	0.965	<i>per-probe</i>	0.942	65%

Encoders with higher effective rank relative to sample size require stronger correction (c further from 1), consistent with the finite-sample upper-bound mechanism: ResNet-50 features ($d=2048$, $r_{\text{eff}} \approx 350$) are spectrally sparser than ViT features ($d=768$, $r_{\text{eff}} \approx 400$), so the two-component spectral mixture approximation incurs less error. A global $c=0.712$ across all vision probes yields $R^2=0.898$, confirming that a single constant provides a strong baseline; per-probe calibration is a refinement, not a necessity.

Figure 4 shows the E1c scatter plot before and after calibration.

M.8 E_diverse: Cross-Architecture Probe Independence

Design. To test whether the Collapse Model’s predictions are robust to the choice of encoder, we evaluate the same 6 dataset pairs (from E1b) using both ResNet-50 and ViT-B/16 as independent probes, giving 12 total configurations.

Results. ResNet-50: $R^2 = 0.557$, Pearson $r = 0.9987$, 100% superadditive. ViT-B/16: $R^2 = -16.1$ (systematic over-prediction), Pearson $r = 0.9825$, 83.3% superadditive. Cross-architecture Pearson $r \geq 0.982$ confirms that *directional rankings* are probe-independent even when absolute magnitudes

838 differ. The Pearson vs. R^2 gap for ViT-B/16 shows encoder-dependent scale bias despite stable
839 ordering.

840 M.9 E4: Cross-Probe Consistency

841 **Design.** We compute the 6×6 pairwise SA_k alignment matrix M independently using ResNet-50
842 and ViT-B/16 on 4 common datasets {CIFAR-10, MNIST, STL-10, SVHN}. The 6 upper-triangle
843 entries of each matrix are compared via Pearson r .

844 **Results.** Cross-probe Pearson $r = 0.891$ ($n = 6$, $p = 0.017$). Both probes agree on the direction
845 of domain distance for all 6 pairs. CIFAR-10/STL-10 are close in both; CIFAR-10/MNIST are far
846 in both. This confirms the SA_k distance matrix reflects intrinsic inter-dataset geometry rather than
847 encoder artifacts.

848 M.10 E_downstream: Effective Rank as Downstream Quality Proxy

849 **Motivation.** The Collapse Model predicts r_{eff} of merged features, but does higher r_{eff} actually
850 correspond to better downstream performance? We test this by computing r_{eff} and linear-probe (LP)
851 accuracy for 10 pretrained encoders on 5 datasets (50 total pairs).

852 **Design.** Encoders: ResNet-{50, 101}, EfficientNet-B4, ConvNeXt-T, DenseNet-121, ViT-B/16,
853 DeiT-B, Swin-T, BEiT-B, RegNet-Y (covering CNN, ViT, and hybrid families). Datasets: CIFAR-10,
854 CIFAR-100, STL-10, SVHN, Fashion-MNIST. For each (encoder, dataset) pair: extract $N=2000$
855 train features, compute r_{eff} ; train LP (sklearn LogisticRegression) on train, evaluate on test. Report
856 per-dataset Spearman $\rho(r_{\text{eff}}, \text{LP acc})$ across 10 encoders.

857 Results.

Dataset	All 10 probes		ViT family ($d=768$)		CNN family ($d \geq 1024$)	
	n	ρ	n	ρ	n	ρ
CIFAR-10	10	+0.64	5	+0.90	5	+0.30
CIFAR-100	10	+0.61	5	+0.90	5	+0.30
STL-10	9	+0.73	5	+0.90	4	+0.20
SVHN	10	-0.03	5	+0.10	5	-0.30
Fashion-MNIST	10	+0.03	5	+0.60	5	-0.10

858 The “all probes” column ($\rho \approx +0.59$ on natural images) is *confounded by feature dimension*: CNNs
859 ($d=2048$) have systematically higher r_{eff} than ViTs ($d=768$), inflating the apparent correlation.
860 Controlling for dimension reveals a sharper picture: **within the ViT family** ($d=768$, 5 probes), r_{eff}
861 is strongly predictive of LP accuracy on natural images ($\rho=0.90$ on CIFAR-10, CIFAR-100, and
862 STL-10), while **within the CNN family** ($d \geq 1024$, 5 probes) the correlation is weak ($\rho=0.20-0.30$).
863 This suggests that r_{eff} captures useful discriminative information primarily for architectures with
864 compact feature spaces ($d < 1024$), where spectral diversity is more tightly coupled to representation
865 quality.
866

867 **Implication for the Collapse Model.** r_{eff} should be understood as a measure of *representational*
868 *diversity*—the number of independent directions activated by the data—rather than a direct proxy
869 for downstream accuracy. The Collapse Model predicts when merging *increases diversity*, which
870 is a necessary (but not sufficient) condition for improving downstream performance. Within an
871 architecture family, diversity and accuracy are positively correlated ($\rho=0.90$ for ViTs), justifying the
872 use of r_{eff} -based selection among candidates extracted from the same encoder. Across architecture
873 families or across datasets of varying difficulty, additional task-specific metrics are needed.

874 **Sample-size sensitivity.** Empirically, the over-prediction ratio grows with N/r_{eff} on 10 representa-
875 tive ResNet-50 pairs: at $N=200$ the mean ratio is 1.021, rising monotonically to 1.545 at $N=2000$
876 (Spearman $\rho=0.80$ between N/r_{eff} and log bias). This association helps predict calibration variation
877 across encoders (ResNet-50 $c=0.78$, DINO $c=0.58$), but is not asserted as a theoretical error law.

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Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

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Justification: The abstract and introduction (Section 1) distinguish four contributions—the exact Gram representation and paired-direction decomposition (Proposition 1, Lemma 2), the structured summary predictor and its diagnostic threshold (Eq. 9, Corollary 4), Collapse-guided greedy compression (Section 5), and cross-modal validation on 427 pairs. Proposition 5 proves that the four summaries are not sufficient for universal exact prediction. The assumptions, empirical scope, calibration dependence, and failure modes are stated explicitly in Appendix A.

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Question: Does the paper discuss the limitations of the work performed by the authors?

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Justification: Appendix A discusses the uniform- α assumption, its failure on cross-domain CLIP/DINO pairs (bimodal principal-angle distribution, Fig. 2), and the degraded absolute-prediction accuracy on SBERT text pairs ($r=0.08$ of explained variance vs. 0.96 for vision). Ranking correctness is preserved in both failure modes. The need for a piecewise model for non-uniform V_α is identified as future work.

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Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

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Justification: Proposition 1, Lemma 2, and Proposition 3 state their assumptions explicitly in Section 3.2. The exact results require, respectively, no additional assumptions, paired and decoupled directions, and proportional spectra with uniform alignment. Equation 9 is identified as an approximation for general spectra, and Proposition 5 gives a counterexample to universal identification from four scalars. Full proofs are provided in Appendix B; all results are numbered and cross-referenced.

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Justification: Algorithm 1 (Appendix L) provides full pseudocode for the Collapse-guided greedy compression procedure. Appendix M lists all hyperparameters ($N=1,000$, bootstrap $B=10,000$, calibration constants). Section 4 specifies all 5 encoder families, 14 vision datasets, and 19 SBERT text corpora, with dataset sizes and evaluation protocols. All datasets are publicly available.

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1035 Justification: Section 4 specifies all 5 encoder families, sample sizes ($N=1,000$), and
1036 evaluation protocols (Pearson r , Spearman ρ , R^2 , sign test). Appendix M provides the
1037 complete hyperparameter table, including bootstrap resamples ($B=10,000$), calibration
1038 constants, and the greedy compression budget k . Per-probe calibration methodology is
1039 described in Appendix L.

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1052 Sign tests (with p -values) are reported for LP accuracy comparisons. The $\Delta\rho=0.003$ gap
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1083 full pipeline requires one GPU-equipped machine; CPU-only machines can reproduce all
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